

72nd BPSC PRELIMS

प्रहार

Test Series

Test-22/6

FULL LENGTH TEST

Solution

English



BPSC TEST-22/6 (SOLUTION)

Q 1. Ans. C

Explanation:

- The river Ganga finally exits the state of Bihar near the town of Manihari in the Katihar district.
- **Entry Point:** The river enters Bihar at Chausa in the Buxar district.
- **Flow:** It acts as the principal drainage channel of the state, flowing from west to east and dividing Bihar into two unequal halves (North and South Bihar).

Q 2. Ans. C

Explanation:

- **Bilasi Dam:** Located in the Banka district.
- **Jalkund Dam:** Located in the Munger district.
- **Kohira Dam:** Located in the Kaimur (Bhabua) district.
- **Barnar Dam:** Located in the Jamui district.

Q 3. Ans. D

Explanation:

- National Waterway-94 (NW-94) is located on the Son River in Bihar.
- Under the National Waterways Act of 2016, which declared 111 inland waterways across the country, a 141-kilometre stretch of the Son River was officially designated as NW-94 to facilitate cargo movement and promote inland water transport in the state.

Q 4. Ans. C

Explanation:

- **Pair 1 is incorrect:** The Bansi River is not a tributary of the Kiul. It is a historically significant river that flows along the UP-Bihar border (near West Champaran) and eventually acts as a tributary to the Narayani (Gandak) River at Pipraghat.
- **Pair 2 is incorrect:** The Bandarbari River is not a tributary of the Gandak. It is a small river/stream located in the Jamui district of South Bihar, making it a part of the Kiul River basin.
- **Pair 3 is correct:** The Baya River is a vital tributary/distributary in North Bihar. It flows for roughly 212 kilometres from East Champaran, passing through the Tirhut region, and ultimately merges into the Ganga River near Begusarai.
- **Pair 4 is correct:** The Ausane River is a minor tributary of the Sone River. It flows through the deep gorges of the Rohtas Plateau in South Bihar.

The famous Rohtasgarh Fort is strategically situated such that it is protected by the Sone River on two sides and the Ausane River on the third.

Q 5. Ans. B

Explanation:

- **Statement 1 is incorrect:** While Abdul Bari was deeply involved with the committee, he was appointed as its Vice-Chairman, not the Chairman. The Bihar Labour Enquiry Committee was actually under the chairmanship of Dr. Rajendra Prasad.
- **Statement 2 is correct:** After engaging in the freedom struggle under leaders like Rajendra Prasad and A.N. Sinha, he shifted significant focus to the labour movements. He organised workers against exploitation at the Tata Iron and Steel Company (Jamshedpur) and adjoining coal mines, eventually being chosen as the President of the Tata Workers' Union.
- **Statement 3 is correct:** Due to his stellar organisational ability and political standing from his time in the Quit India movement (during which he was imprisoned in Hazaribagh Jail), he was elevated to the position of President of the Bihar Pradesh Congress Committee in 1946. He held this post until he was tragically assassinated on the Patna-Fatuha road on 28 March 1947 while en route to meet Mahatma Gandhi.

Q 6. Ans. B

Explanation:

- **Statement 1 is correct:** The Khandavala dynasty (which ruled the Darbhanga Raj) was founded by the Maithil Brahmin scholar Mahesh Thakur in the 16th century. He was granted the territory of Mithila by Mughal Emperor Akbar as a reward for his vast scholarship and intellect.
- **Statement 2 is incorrect:** Despite its massive wealth, standing army, and the rulers holding the title of Maharajadhiraja, the Darbhanga Raj was technically a **Zamindari estate**, not a recognised, sovereign "Princely State" (like Mysore or Hyderabad). They did not have a gun salute recognised by the British Crown. It was, however, the largest and richest zamindari in India.
- **Statement 3 is correct:** The rulers of Darbhanga were massive patrons of education. Maharaja Rameshwar Singh donated huge sums of money

(amounting to lakhs of rupees at the time) to Pandit Madan Mohan Malaviya for the establishment of the Banaras Hindu University (BHU). They also heavily funded Patna University, Calcutta University, and established the Kameshwar Singh Darbhanga Sanskrit University.

- **Statement 4 is correct:** The Darbhanga Raj court was a major cultural epicentre. They provided immense patronage to musicians, heavily supporting the ancient Dhrupad style of Hindustani classical music, which resulted in the establishment and popularisation of the renowned **Darbhangha Gharana**.

Q 7. Ans. D

Explanation:

- **Statement 1 is correct:** The primary objective of the Bihar Jan Vishwas (Amendment of Provisions) Act, 2026, is to promote trust-based governance. By decriminalising minor, technical, and procedural defaults, the state government aims to significantly improve the “Ease of Doing Business” and the “Ease of Living” for its citizens and entrepreneurs.
- **Statement 2 is correct:** The Act amends multiple state legislations (such as the Bihar Municipal Act, 2007, and the Societies Registration Act, 1860) to remove the fear of criminal prosecution. It explicitly replaces the provisions of imprisonment for minor offences with purely administrative and monetary penalties.
- **Statement 3 is correct:** To ensure that the newly implemented financial fines do not lose their deterrent effect over time due to inflation, the Act introduces a periodic escalation clause. It mandates that the minimum amount of fine or penalty will automatically increase by 10% after the expiry of every three years.

Q 8. Ans. B

Explanation:

- **Suvidha Vahini Portal:** Associated with the **Bihar Police** to streamline citizen-centric police services and ease the processes of verification and clearance.
- **e-Amrit App / Platform:** Implemented by the **Health Department** as a comprehensive digital health platform to connect beneficiaries, health facilities, and frontline workers (like ASHAs) to ensure a seamless healthcare continuum in the state. (Note: Do not confuse this with NITI Aayog’s national “e-Amrit” portal for Electric Vehicles).
- **BEFQR App:** Launched by the **Water Resources Department** of Bihar. It is used to monitor and

mitigate the impacts of floods and droughts in the state, ensuring safe and smooth management of water resources.

- **e-Nishchay Portal:** Launched by the **Panchayati Raj Department**. This portal (and its accompanying mobile app) was created so citizens can easily register and track online complaints regarding irregularities in the Har Ghar Nal Ka Jal (tap water) and Gali-Nali Pakkiran (paved streets and drains) schemes under the state’s Saat Nishchay initiative.

Q 9. Ans. C

Explanation:

- Jitwarpur village in Madhubani district has become Bihar’s first Craft Village.
- **The village is associated with Mithila painting and has produced three Padma Shri awardees:** Jagadamba Devi, Sita Devi and Baua Devi.
- Ranti (Madhubani), Raiyam (Darbhanga), Bhusra (Muzaffarpur) and Tharkatti (Gaya) are also being developed as Craft Villages.
- This initiative, developed under the National Handicrafts Development Programme, aims to preserve the living tradition of Mithila (Madhubani) art and provide a dedicated hub for artisans to practice, display, and sell their work globally.
- **Upcoming Craft Villages:** The Bihar Government and Upendra Maharathi Shilp Anusandhan Sansthan are also developing craft villages in:
 - ✓ Ranti (Madhubani)
 - ✓ Raiyam (Darbhanga)
 - ✓ **Bhusra (Muzaffarpur):** Famous for Sujani embroidery.
 - ✓ **Tharkatti (Gaya):** Famous for stone carving.

Q 10. Ans. C

Explanation:

- **Sailesh (Shailesh) Kumar**, a prominent para-athlete hailing from Islamnagar village in the **Jamui** district of Bihar, recently won his maiden gold medal at the World Para Athletics Championships held at the Jawaharlal Nehru (JLN) Stadium in New Delhi by clearing a personal best height of 1.91m.
- Sailesh defeated the reigning Paralympic champion and world-record holder, USA’s Ezra Frech. Fellow Indian Varun Singh Bhati took the bronze.

Q 11. Ans. B

Explanation:

- A new National Institute of Electronics and Information Technology (NIELIT) Centre was recently inaugurated in **Muzaffarpur**, Bihar.

- In October 2025, Union Minister for Electronics and Information Technology, Ashwini Vaishnaw, virtually inaugurated this facility alongside four other new centres across India. The establishment of this centre in Muzaffarpur aims to democratize access to high-quality digital education and provide youth in the region with industry-focused training in niche emerging technologies like Artificial Intelligence, Data Science, Cybersecurity, and Semiconductors.

Q 12. Ans. D

Explanation:

- Bihar's first Manuscript Cluster Centre has been established at the Nav Nalanda Mahavihara in Nalanda.
- The Union Ministry of Culture recently designated Nav Nalanda Mahavihara as the state's first cluster centre under its flagship 'Gyan Bharatam' initiative.
- The centre is tasked with identifying, digitising, and conserving rare and ancient manuscripts scattered across the state, thereby preserving Bihar's rich literary and historical heritage for future generations.
- Khuda Bakhsh Oriental Public Library in Patna is also a part of this initiative, but it was onboarded as an 'Independent Centre', whereas Nav Nalanda Mahavihara is the designated 'Cluster Centre' for the state.

Q 13. Ans. B

Explanation:

- Senior BJP leader **Dr. Prem Kumar** was unanimously elected as the Speaker of the newly formed 18th Bihar Legislative Assembly on December 2, 2025.
- The election was officially announced by Pro-tem Speaker Narendra Narayan Yadav during the very first sitting of the 18th Assembly.

Q 14. Ans. C

Explanation:

- Bihar Chief Minister Nitish Kumar was officially inducted into the World Book of Records, London, an acknowledgement of a rare milestone in democratic politics: taking the oath of office for a record **10th time** on November 20, 2025.
- Having first assumed the post for a brief seven-day stint on March 3, 2000, his cumulative time in office exceeds 19 years and 3 months.
- This establishes him as the longest-serving Chief Minister in Bihar's history.

Q 15. Ans. C

Explanation:

- **Statement 1 is correct:** Babu Kunwar Singh was born in 1777. Following the death of his father, Sahibzada Singh, in **1826**, he inherited the Jagdishpur estate (in present-day Bhojpur district, Bihar).
- **Statement 2 is correct:** The Indian sepoys stationed at the Danapur Cantonment mutinied on **25 July 1857**. They crossed the Son River, marched into Arrah, and officially accepted the 80-year-old Kunwar Singh as their leader, formally igniting the rebellion in Bihar.
- **Statement 3 is incorrect:** Kunwar Singh reached Lucknow after visiting Rewa, Banda, and Kalpi, not before. His military campaign followed a distinct westward trajectory before looping back to Bihar.
- **Statement 4 is correct:** While retreating toward Bihar after the Battle of Azamgarh, Kunwar Singh crossed the Ganga at **Shivpur Ghat** (near Ghazipur) on **21 April 1858**. During this crossing, he was ambushed by British forces led by Brigadier Douglas. A bullet shattered his left wrist, prompting him to sever his own hand with a sword and offer it to the river to prevent infection. He reached Jagdishpur the next day.

Q 16. Ans. D

Explanation:

- The Vidatha is regarded as the oldest tribal assembly mentioned in the Rig Veda. It was a multifunctional institution that performed military, religious, and secular functions.
- **Features of the Vidatha:**
 - ✓ Discussed matters related to warfare and military organisation.
 - ✓ Conducted religious ceremonies and sacrifices.
 - ✓ Served as a gathering for social and economic activities.
 - ✓ Both men and women are believed to have participated in its proceedings.

Q 17. Ans. C

Explanation:

- The early Jain teachers and Tirthankaras, especially Mahavira, preached their doctrines in Prakrit, particularly Ardhamagadhi Prakrit, so that their teachings could be easily understood by the common people.
- At a time when Sanskrit was largely confined to the educated and priestly classes, Jainism adopted the vernacular language of the masses, which helped in the wider spread of its teachings.

Q 18. Ans. C**Explanation:**

- The Kushanas adopted the Satrapy system of provincial administration, a practice influenced by earlier Indo-Greek and Shaka traditions. Under this system, provinces were governed by officials known as Satraps (Kshatrapas) and Mahakshatrapas (Great Satraps).
- This administrative arrangement was mainly confined to the northern and western regions of India, which formed the core areas of the Kushana Empire. These governors exercised considerable authority while remaining subordinate to the Kushana ruler.

Q 19. Ans. A**Explanation:**

- **Statement 1 is correct** : Swami Vivekananda regarded Vedanta as a highly rational and universal philosophical system. He believed that Vedanta provided a logical basis for understanding the unity of existence and harmonizing religion with reason. Through his speeches and writings, he sought to reinterpret Hindu philosophy in a form suitable for the modern age.
- **Statement 2 is incorrect**: Vivekananda did not advocate isolation from foreign influences. Instead, he emphasised learning from other cultures while remaining firmly rooted in Indian traditions.

Q 20. Ans. C**Explanation:**

- **Kandahar Inscription — Greek and Aramaic bilingual scripts**
 - ✓ The Kandahar inscriptions (Afghanistan) are among Ashoka's most remarkable inscriptions. They were issued in Greek and Aramaic, reflecting the presence of Hellenistic and West Asian populations in the region.
- **Shahbazgarhi Rock Edict — Distinct Kharoshthi script execution**
 - ✓ The Shahbazgarhi Major Rock Edicts (present-day Pakistan) are inscribed in the Kharoshthi script, which was written from right to left and was widely used in the north-western part of the Mauryan Empire.
- **Maski Minor Rock Edict — Explicitly mentions the personal name Ashoka**
 - ✓ The Maski Minor Rock Edict (Karnataka) is historically significant because it is one of the

first inscriptions where “Ashoka” is explicitly mentioned. Most other edicts refer to him as “Devanampiya Piyadasi” (Beloved of the Gods, Pleasant to Behold).

Q 21. Ans. A**Explanation:**

- Akshapatadika maintained and supervised the state's accounts and records.
- Shulkadhyaksha was responsible for collecting customs duties, tolls, and taxes on trade.
- Lakshanadhyaksha supervised coinage and the royal mint, ensuring proper standards of currency.
- Sannidhata was the chief custodian of the treasury and state storehouses.

Q 22. Ans. A**Explanation:**

- **Assertion (A) is true**: During the Later Vedic Period (c. 1000–600 BCE), the importance of early tribal institutions such as the Vidatha gradually declined. The Vidatha, which had performed military, religious, and social functions in the Rig Vedic age, became less significant as society evolved from a tribal to a territorial political structure.
- **Reason (R) is true**: The Later Vedic period witnessed the emergence of larger janapadas and territorial kingdoms. As the king's authority expanded, the political system became more centralised and monarchical. This strengthened the position of the king and reduced the role of popular tribal assemblies such as the Vidatha, Sabha, and Samiti.
- **Why is (R) the Correct Explanation of (A)**: The decline of the Vidatha was directly linked to the rise of larger kingdoms and the increasing concentration of power in the hands of the king. As governance became more centralised, tribal assemblies lost many of their earlier functions.

Q 23. Ans. D**Explanation:**

- In the Mughal revenue system, Khalisa referred to the crown lands whose revenue was collected directly on behalf of the emperor and deposited into the imperial treasury. These lands were administered by officials of the central government, and the income was used to meet the expenses of the imperial court and administration.
- Why Khalisa was important
 - ✓ Revenue went directly to the Mughal state, not to individual nobles.

- ✓ It provided the emperor with an independent source of income.
- ✓ The extent of Khalisa's lands often increased under rulers seeking greater central control over finances.

Q 24. Ans. B

Explanation:

- The Sukkur and Rohri hills in Sindh (Pakistan) served as the principal Harappan factory centre for the extraction, manufacture, and distribution of chert blades. Chert, a hard siliceous stone, was extensively used by the Harappans to make blades, sickles, knives, drills, and other tools.
- **Archaeological evidence shows the presence of:**
 - ✓ Large-scale quarries
 - ✓ Workshops for blade production
 - ✓ Massive quantities of debitage (waste flakes)
- This indicates that the region functioned as a specialised industrial centre supplying chert blades to various Harappan settlements across the Indus civilisation.

Q 25. Ans. A

Explanation:

- **Statement 1 is correct:** The Neolithic site of Chirand, located in Saran district of Bihar on the banks of the Ghaghara River, has yielded a rich collection of bone tools, particularly those made from deer antlers. Excavations have revealed antler-made points, awls, needles, and other implements, indicating a highly developed bone-tool industry. Chirand is one of the most important Neolithic sites of eastern India.
- **Statement 2 is incorrect:** Although Chirand is situated in the alluvial plains of the Ganga basin, the inhabitants did not lack stone tools. Excavations have yielded numerous ground and polished stone celts, axes, chisels, hammer stones, and microliths. Since suitable stone was not locally available, the raw material was likely procured through exchange networks from nearby hilly regions.

Q 26. Ans. B

Explanation:

- **Jupiter** is unequivocally the largest planet in our solar system, both in terms of mass and volume.
- **Size and Volume:** Jupiter's radius is about 11 times that of Earth. To put its massive volume into perspective, you could fit approximately 1,300 Earths inside Jupiter.

- **Mass:** It is a true heavyweight. Jupiter is more than twice as massive as all the other planets in the solar system combined (about 318 times the mass of Earth).
- **Composition:** As a "gas giant," it lacks a true solid surface and is composed primarily of hydrogen and helium, similar to the composition of the Sun.

Q 27. Ans. A

Explanation:

- **Statement 1 is correct:** The Earth spins on its axis from west to east. This movement is called **rotation**, and it takes approximately 24 hours to complete. Because the Earth is spherical, only the half facing the Sun receives light (Day) while the other half is in shadow (Night).
- **Statement 2 is correct:** The Earth travels around the Sun in an elliptical orbit, a movement known as **revolution**. This revolution, combined with the fact that Earth's axis is tilted at an angle of 23.5 degrees, causes different hemispheres to receive varying angles of direct sunlight throughout the year. This combination of revolution and axial tilt creates the **seasons**.
- **Statement 3 is incorrect:** The Earth does not take exactly 365 days to orbit the Sun. A complete revolution takes **365 days and approximately 6 hours** (365.2422 days, to be precise). To keep our calendars aligned with the astronomical seasons, we count a standard year as 365 days and accumulate the leftover 6 hours. Every four years, those hours add up to one full day (24 hours), which is added to the month of February, creating a **Leap Year**.

Q 28. Ans. C

Explanation:

- The **Inner Core** is the absolute center and innermost layer of the Earth. Despite temperatures that rival the surface of the Sun (around 5,400°C or 9,800°F), it is a dense, solid ball composed primarily of iron and nickel. It remains solid because the immense pressure at the center of the Earth prevents the metals from melting.

Q 29. Ans. B

Explanation:

- A braided channel consists of a network of small water channels separated by small, often temporary, islands called braid bars (or mid-channel bars).
- This formation occurs when a river experiences a significant decrease in velocity or flow volume,

causing it to lose the energy required to transport its sediment load.

- The river deposits its heavy sediment (usually coarse sand and gravel) directly into the middle of its own channel, forming a central bar.
- Because the path is now blocked by this newly deposited sediment, the water is forced to divert and flow around the bar, splitting the main river into multiple interweaving, shallower channels.
- Classic examples of braided rivers in India include the Kosi (in north Bihar, before its embanked stretches), the Brahmaputra (in Assam), the upper reaches of many Himalayan rivers (Ganga, Yamuna, Sutlej, Beas in the foothills), and many glacial-fed streams.

Q 30. Ans. D

Explanation:

- The permanent winds that blow from the Subtropical High-Pressure Belts (located around 30° N and 30° S latitudes) towards the Equatorial Low-Pressure Belt (near the equator) are called the Trade Winds.
- Because the Earth rotates, the Coriolis force deflects these winds. Instead of blowing straight north to south, they are deflected to the right in the Northern Hemisphere (becoming the Northeast Trade Winds) and to the left in the Southern Hemisphere (becoming the Southeast Trade Winds).
- The term “trade” originally comes from an old

Middle English word meaning “path” or “track,” because these winds blow so reliably in a constant direction. Historically, sailors and merchants heavily relied on them to cross the oceans and establish global maritime trade routes.

Q 31. Ans. A

Explanation:

- **Statement 1 is correct:** The **Troposphere** is the lowermost layer of Earth’s atmosphere. It begins at the Earth’s surface and extends upward to an average height of about 13 km (roughly 8 km at the poles and 18 km at the equator).
- **Statement 2 is correct:** The **Ozone layer** is predominantly located within the lower portion of the **Stratosphere**, approximately 15 to 35 kilometers above Earth. This crucial layer absorbs the majority of the Sun’s harmful ultraviolet (UV) radiation.
- **Statement 3 is incorrect:** Almost all weather phenomena, such as clouds, rain, storms, and fog, occur in the **Troposphere**, not the stratosphere. The troposphere contains about 75% of the atmosphere’s mass and 99% of its water vapor and aerosols. The stratosphere, by contrast, is very dry and stable, which is why commercial jet aircraft often fly in the lower stratosphere to avoid weather-related turbulence.

Q 32. Ans. B

Explanation:

Mountain Range	Continent	Approximate Length	Key Distinction
Andes	South America	~7,000 km	The longest continental mountain range.
Rockies	North America	~4,800 km	The major mountain system of western North America.
Himalayas	Asia	~2,400 km	Home to the world's highest peaks.
Alps	Europe	~1,200 km	The highest and most extensive mountain range is located entirely in Europe.

Q 33. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The Amur River (known as the Heilong Jiang in Chinese) forms a massive, roughly 2,800-kilometer natural boundary between the Russian Far East and Northeastern China.
- **Pair 2 is incorrectly matched:** While the Limpopo River does form a northern border for South Africa, it separates South Africa from Botswana and

Zimbabwe, not Namibia. The river that forms the border between South Africa and Namibia is the Orange River.

- **Pair 3 is correctly matched:** The Mekong River acts as a major natural geographical boundary in Southeast Asia, separating Laos and Thailand for hundreds of kilometers before flowing into Cambodia and Vietnam.
- **Pair 4 is incorrectly matched:** The Oder River forms the international border between Germany and Poland (known as the Oder-Neisse line), not

France. The major river that forms the natural border between Germany and France is the Rhine.

Q 34. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The Isthmus of Kra is the narrow neck of southern Thailand that acts as a land bridge, connecting the Malay Peninsula to mainland Asia.
- **Pair 2 is incorrectly matched:** The Isthmus of Corinth does not connect Italy to Europe; it connects the Peloponnese peninsula to mainland Greece. (The Italian peninsula is connected to mainland Europe by the Alps mountain range, not an isthmus.)
- **Pair 3 is incorrectly matched:** The Isthmus of Suez connects the continents of Africa and Asia, but it separates the Mediterranean Sea and the Red Sea, not the Black Sea. The bodies of water separating the Mediterranean/Marmara from the Black Sea are the Bosphorus and Dardanelles straits.
- **Pair 4 is correctly matched:** The Isthmus of Panama is the famous narrow strip of land linking North and South America, effectively acting as a barrier that separates the Caribbean Sea (Atlantic Ocean) and the Pacific Ocean.

Q 35. Ans. B

Explanation:

- **Statement 1 is correct:** The Pranhita River is the largest tributary of the Godavari. It is formed by the confluence of the Wardha and Wainganga rivers (which themselves carry the waters of the Penganga). It joins the Godavari at Kaleshwaram in Telangana, a site famous for the massive Kaleshwaram Lift Irrigation Project and the Kaleshwara Mukteswara Swamy Temple.
- **Statement 2 is incorrect:** The statement reverses the geographical order. Descending from the upper Himalayas, the Alaknanda and Pindar rivers meet at Karnaprayag. It is only after this point, further downstream at Rudraprayag, that the Alaknanda and Mandakini rivers meet. The correct descending sequence of the Panch Prayag is: Vishnuprayag → Nandaprayag → Karnaprayag → Rudraprayag → Devprayag.
- **Statement 3 is incorrect:** The Harike Wetland (Harike Pattan) is one of the most important wildlife sanctuaries in Northern India, but it is situated at the confluence of the Sutlej and the Beas rivers, not the Ravi.
- **Statement 4 is correct:** The Tungabhadra River

(formed by the Tunga and Bhadra rivers) flows through Karnataka and Andhra Pradesh before joining the Krishna River at Sangameswaram in the Nandyal district of Andhra Pradesh. The site is known for the submerged Sangameswara Temple, which surfaces only during the dry season.

Q 36. Ans. D

Explanation:

- Gugamal National Park is located in the Amravati district of Maharashtra, in the Vidarbha region of the state. It was established in 1975 and covers an area of about 361.28 sq km. The park is part of the larger Melghat Tiger Reserve, which is one of India's earliest Project Tiger reserves (notified in 1973–74 among the first nine tiger reserves). The park lies in the Satpura Range and the Gawilgarh Hills, and the area is drained by tributaries of the Tapi River. It is known for its sal and teak forests and supports a wide range of wildlife, including tigers, leopards, sloth bears, sambar, gaur, and the rare wild dog (dhole).
- Gujarat is associated with parks like Gir National Park (Asiatic lions), Blackbuck National Park (Velavadar), and Marine National Park (Gulf of Kachchh).
- Chhattisgarh has Indravati, Kanger Valley, and Guru Ghasidas National Parks.
- Rajasthan has Ranthambore, Sariska, Keoladeo (Bharatpur), Desert National Park, and Mukundra Hills.
- **Maharashtra has six national parks:** Gugamal, Tadoba, Sanjay Gandhi (Borivali), Chandoli, Pench (Maharashtra portion), and Navegaon. Gugamal sits within the Melghat Tiger Reserve, a critical tiger habitat that also includes the adjoining Melghat Wildlife Sanctuary.

Q 37. Ans. B

Explanation:

- **Statement 1 is incorrect:** While the European Union's Carbon Border Adjustment Mechanism (CBAM) entered its definitive compliance phase in 2026, meaning importers must now surrender paid certificates, it does not completely ban "default values." The European Commission still legally permits the use of conservative default values as a strict fallback if actual data is unavailable, and up to 20% for "complex goods."
- **Statement 2 is correct:** Under the Paris Agreement, Article 6.4 creates a centralised global carbon market

governed by a UN Supervisory Body. This mechanism (the Paris Agreement Crediting Mechanism, or PACM) is the direct, modernised successor designed to replace the outdated Clean Development Mechanism (CDM) of the Kyoto Protocol.

Q 38. Ans. C

Explanation:

- **Pair 1 is correctly matched:** The **Sahara** is the largest hot desert in the world, spanning across vast swathes of **North Africa**.
- **Pair 2 is correctly matched:** The **Atacama** Desert is a coastal desert located on the western seaboard of **South America**, primarily in northern Chile. It is renowned for being one of the driest non-polar deserts on Earth.
- **Pair 3 is correctly matched:** The **Gobi** Desert is a vast, cold winter desert region in **East Asia**, covering parts of northern China and southern Mongolia.

Q 39. Ans. A

Explanation:

- **Statement 1 is correct:** The National Flag in its present form was adopted by the Constituent Assembly on 22 July 1947, before the country became independent, and its design and significance were explained to the Assembly at that time.
- **Statement 2 is correct:** The Constituent Assembly adopted Jana Gana Mana as the National Anthem and gave Vande Mataram an equal status of honour as the National Song on the same day, 24 January 1950, two days before the Constitution came into force.
- **Statement 3 is incorrect:** The Constitution itself does not contain any provision declaring Jana Gana Mana to be the National Anthem; the anthem and the song were adopted by the Constituent Assembly through a statement made by its President. The Assembly had earlier been constituted under the Cabinet Mission Plan and had adopted the Objectives Resolution moved by Jawaharlal Nehru.

Q 40. Ans. A

Explanation:

- **Statement 1 is correct:** The territory of India is wider in scope than the Union of States, since the Union of States includes only the states that share a distribution of powers with the Centre, whereas the territory of India comprises the states, the union territories, and any other territories that may be acquired by India.

- **Statement 2 is incorrect:** The states in India do not enjoy any right to secede from the Union; the country is described as an indestructible union of destructible states, meaning that while the boundaries and identities of states can be altered by Parliament, the Union itself cannot be broken up by any state leaving it.

- Article 1 describes India as a Union of States, and the term Union was preferred to Federation precisely to convey that the Indian federation is not the result of an agreement among the states and that no state has the right to secede. The evolution of states and union territories has proceeded through reorganisations beginning with the States Reorganisation Act of 1956.

Q 41. Ans. A

Explanation:

- **Habeas Corpus:** The writ of habeas corpus, meaning to have the body, is an order to produce the body of a detained person before the court so that the legality of the detention can be examined; it can be issued against both public authorities and private individuals.
- **Mandamus:** The writ of mandamus, meaning we command, is issued to a public official, a public body, or a lower court to compel the performance of an official duty that it has failed to perform; it cannot be issued against a private individual, against the President or Governors, or to enforce a discretionary duty.
- **Quo Warranto:** The writ of quo warranto, meaning by what authority, is issued to inquire into the legality of a person's claim to a substantive public office, and it can be sought by any interested person, not necessarily the aggrieved party.
- **Certiorari:** The writ of certiorari, meaning to be certified, is issued by a higher court to a lower court or tribunal to quash an order already passed, on grounds such as excess of jurisdiction or error of law. The fifth writ, prohibition, is issued to prevent a lower court from exceeding its jurisdiction while proceedings are still pending.

Q 42. Ans. A

Explanation:

- **Statement 1 is correct:** The right to property is now a legal or constitutional right under Article 300A, which provides that no person shall be deprived of his property save by authority of law; it was removed from the list of fundamental rights by

the 44th Amendment Act of 1978, which deleted Article 19(1)(f) and Article 31.

- **Statement 2 is incorrect:** Since the right to property is no longer a fundamental right, its violation cannot be challenged directly before the Supreme Court under Article 32, which is available only for the enforcement of fundamental rights; an aggrieved person can instead approach a High Court under Article 226 or pursue ordinary legal remedies.
- The deletion of the fundamental right to property was intended to remove the obstacles that the right had placed in the way of land reforms and socioeconomic legislation. As a legal right, the right to property can be regulated or taken away by an ordinary law of the legislature, without the need for a constitutional amendment, subject only to the requirement that there be authority of law.

Q 43. Ans. B

Explanation:

- **Statement 1 is correct:** The creation of a new state, or the alteration of the area, boundary, or name of an existing state under Article 3, can be effected by a simple majority of Parliament, and such a law is expressly stated not to be a constitutional amendment for the purposes of Article 368.
- **Statement 2 is correct:** Amending the provisions relating to the distribution of legislative powers between the Centre and the states requires, in addition to a special majority of Parliament, ratification by the legislatures of not less than one-half of the states, because such provisions affect the federal structure.
- **Statement 3 is incorrect:** Not every constitutional amendment requires ratification by the states; ratification is required only for amendments affecting specified federal matters, such as the election of the President, the extent of the executive power of the Union and the states, the Supreme Court and the High Courts, the representation of states in Parliament, and Article 368 itself. Most amendments are passed by a special majority of Parliament alone, and certain matters can be amended by a simple majority outside Article 368.

Q 44. Ans. A

Explanation:

- **Statement 1 is correct:** The President is the supreme commander of the defence forces of India, and the appointments of the chiefs of the army, the navy, and the air force are made in his name, though the

exercise of this military power is regulated by law made by Parliament.

- **Statement 2 is correct:** The President appoints the Chief Justice and the other judges of the Supreme Court, as well as the judges of the High Courts, acting on the recommendation of the collegium, and he also appoints the Governors, the Attorney General, the Comptroller and Auditor General, and the heads of various commissions.
- **Statement 3 is incorrect:** The President cannot declare war or conclude peace independently of Parliament; although treaties and agreements are negotiated and concluded in his name, the exercise of his diplomatic and military powers is subject to the approval of Parliament. The President also enjoys legislative, financial, judicial, and emergency powers, all exercised on the advice of the Council of Ministers.

Q 45. Ans. B

Explanation:

- **Central Information Commission:** Established under the Right to Information (RTI) Act, 2005. Its nodal administrative ministry is the **Ministry of Personnel, Public Grievances and Pensions** (specifically the Department of Personnel and Training - DoPT).
- **National Human Rights Commission:** Established under the Protection of Human Rights Act, 1993. It falls under the administrative purview of the **Ministry of Home Affairs**.
- **Competition Commission of India:** Established under the Competition Act, 2002, to promote and sustain market competition and protect consumer interests. It functions under the **Ministry of Corporate Affairs**.
- **Unique Identification Authority of India:** Originally set up as an attached office of the Planning Commission, it was later established as a statutory body under the Aadhaar Act, 2016. It now operates under the **Ministry of Electronics and Information Technology (MeitY)**.

Q 46. Ans. A

Explanation:

- The voting age was reduced from twenty-one to eighteen years by the 61st Constitutional Amendment Act of 1988, which amended Article 326 in order to provide unrepresented youth with an opportunity to express their feelings and to channel their energies into the political process.

- Universal adult franchise is not a directive principle; it is provided under Article 326 as the basis for elections to the Lok Sabha and the state legislative assemblies, and it is regarded as one of the salient features of the Constitution and a bold experiment in view of the size of the country and the extent of poverty and illiteracy.
- Universal adult franchise means that every citizen who is not less than eighteen years of age has the right to vote without any discrimination based on caste, race, religion, sex, or property, subject to disqualifications such as non-residence, unsoundness of mind, crime, or corrupt and illegal practices. It makes democracy broad-based and enhances the self-respect and prestige of the common people.

Q 47. Ans. C

Explanation:

- **Assertion is true:** Indian courts, in particular the Supreme Court and the High Courts, can declare a law unconstitutional if it contravenes the provisions of the Constitution, especially the fundamental rights, or if it is enacted beyond the competence of the legislature concerned.
- **Reason is false:** The Constitution does not expressly use the term judicial review anywhere; the power of judicial review is implied from several provisions, such as Article 13, which declares laws inconsistent with the fundamental rights to be void, and Articles 32, 131 to 136, 143, 226, 246, 251, 254, and 372.
- Although narrower in scope than in the United States because of the use of the expression procedure established by law, the power of judicial review is a basic feature of the Constitution and cannot be taken away even by a constitutional amendment, as held in the *Minerva Mills* case. It is exercised in respect of both legislative enactments and executive orders of the Centre and the states.

Q 48. Ans. A

Explanation:

- **Statement 1 is incorrect:** While it is true that the Calling Attention Motion is an Indian innovation (introduced in 1954), it is **explicitly mentioned** in the Rules of Procedure. Zero Hour, which is also an Indian innovation, is an informal device not mentioned in the Rules of Procedure.
- **Statement 2 is incorrect:** Amendments to the Motion of Thanks can be moved in both the Lok Sabha and the Rajya Sabha. While the passing of an

amendment in the Lok Sabha essentially amounts to a defeat of the government (as it shows a lack of majority), an amendment passed in the Rajya Sabha (which has happened multiple times, such as in 2015 and 2016) is just a severe political embarrassment, as the government does not need a majority in the Upper House to survive.

- **Statement 3 is correct:** The primary purpose of an Adjournment Motion is to draw the attention of the House to a definite matter of urgent public importance. However, because its adoption implies a strong censure (condemnation) of the government's handling of that specific matter, this power is restricted solely to the Lok Sabha. The Rajya Sabha is not permitted to make use of this device.

Q 49. Ans. D

Explanation:

- **Statement 1 is correct:** The right to move the Supreme Court under Article 32 for the enforcement of Fundamental Rights is itself a Fundamental Right, making the SC's writ jurisdiction mandatory. Conversely, Article 226 grants High Courts a wider scope ("any other purpose"), but the court retains the discretion to decline a writ if alternative remedies are efficacious.
- **Statement 2 is correct:** In *Daryao v. State of U.P.* (1961), the Supreme Court held that if a High Court dismisses a writ petition under Article 226 on merits, after a full hearing, the principle of *res judicata* applies and bars the petitioner from filing a fresh petition on the same cause of action under Article 32. However, if the earlier dismissal was on technical grounds, such as default, withdrawal, or without examining the merits, *res judicata* does not apply, and the petitioner retains the right to approach the Supreme Court directly under Article 32.
- **Statement 3 is correct:** Article 136 (Special Leave Petition) is a plenary, discretionary power of the Supreme Court. It overrides any statutory provisions that attempt to oust the jurisdiction of courts. Even in the presence of an "exclusion of jurisdiction" clause in a statute, the SC can intervene via Article 136 to prevent miscarriage of justice.

Q 50. Ans. A

Explanation:

- **Dilatory Motion:** This is a procedural device moved by a member to supersede or postpone the debate or discussion on a bill or any other motion. It effectively stalls the progress of a specific business.

- **Closure Motion:** A motion moved by a member to cut short the debate on a matter before the House. If the motion is approved, the debate is stopped immediately, and the matter is put to a vote.
- **Guillotine:** A specific form of closure motion where, due to time constraints, the Speaker puts all outstanding matters (even those not discussed) to a vote without further debate.
- **Censure Motion:** Moved to express severe disapproval of the policies or actions of the government (or a specific minister). Unlike the No-Confidence Motion, it must specify the reasons for the censure.

Q 51. Ans. C

Explanation:

- **Statement 1 is correct:** Evaporation is a surface phenomenon in which particles of a liquid gain sufficient energy and escape into the vapour state. During evaporation, the particles with higher kinetic energy leave the liquid surface. As a result, the average kinetic energy of the remaining particles decreases, causing the temperature of the liquid to fall and producing a cooling effect.
- **Statement 2 is incorrect:** Evaporation and boiling are different processes. Evaporation can occur at all temperatures below the boiling point, whereas boiling occurs only at a fixed temperature known as the boiling point. Evaporation is a surface phenomenon and takes place slowly, while boiling is a bulk phenomenon involving the entire liquid. Therefore, evaporation does not require the liquid to reach its boiling point.
- **Statement 3 is correct:** Wind speed is one of the factors affecting evaporation. As wind speed increases, the water vapour particles near the liquid surface are carried away more rapidly. This reduces the amount of vapour in the surrounding air and allows more liquid particles to escape into the vapour state. Consequently, the rate of evaporation increases. This is why clothes dry faster on a windy day.

Q 52. Ans. A

Explanation:

- **Crystallisation is a method used to obtain pure solid crystals from a solution.** The process relies on the fact that the solubility of most solids changes with temperature. When a hot saturated solution is cooled, the dissolved substance separates out in the form of crystals while impurities remain dissolved

in the solution. Thus, **crystallisation is based on differences in solubility.**

- Chromatography is used to separate components of a mixture, such as different dyes present in ink. The separation occurs because different substances have different degrees of attraction towards the stationary phase and different solubilities in the moving solvent. As a result, they travel at different rates and get separated. Therefore, **chromatography is based on differences in adsorption.**
- Distillation is used to separate mixtures of liquids or to obtain a pure liquid from a solution. The method depends on differences in the **boiling points** of the components. The liquid with the lower boiling point vaporises first, and the vapour is then condensed to obtain the pure liquid. **Distillation is used in the preparation of distilled water and in separating liquid mixtures having sufficiently different boiling points.**
- Sublimation is the process in which a substance changes directly from the solid state to the gaseous state without passing through the liquid state. Substances such as camphor, ammonium chloride, and naphthalene undergo sublimation. This property is used to separate sublimable solids from mixtures. Therefore, **sublimation is based on the direct conversion of a solid into a gas.**

Q 53. Ans. B

Explanation:

- **Statement 1 is incorrect:** Alpha particles consist of helium nuclei containing two protons and two neutrons, making them relatively massive. Beta particles are electrons or positrons emitted from the nucleus and have a much smaller mass compared to alpha particles. **Therefore, beta rays are much lighter than alpha rays. Because of their greater mass, alpha particles have lower penetrating power but produce strong ionisation.**
- **Statement 2 is correct:** Gamma rays are electromagnetic radiations emitted from the nucleus during radioactive decay. Unlike alpha and beta particles, gamma rays have **no mass and no charge**. As a result, they possess the greatest penetrating power among the three types of radioactive emissions. While alpha particles can be stopped by a sheet of paper and beta particles by a thin metal sheet, gamma rays require thick layers of lead or concrete for effective shielding.
- **Statement 3 is correct:** Alpha particles carry a **positive charge (+2e)** because they contain two

protons. Since charged particles experience a force in magnetic and electric fields, alpha rays are deflected when they pass through a magnetic field. However, due to their relatively large mass, the deflection is smaller than that of beta particles. Gamma rays, having no charge, are not deflected by magnetic or electric fields.

Q 54. Ans. B

Explanation:

- **Ionisation energy is the minimum amount of energy required to remove the most loosely bound electron from an isolated gaseous atom.** As we move from left to right across a period, the nuclear charge increases while electrons are added to the same shell. The atomic radius decreases, and the attraction between the nucleus and the outermost electrons becomes stronger. Consequently, more energy is required to remove an electron, so **ionisation energy generally increases across a period.**

Q 55. Ans. D

Explanation:

- **Statement 1 is correct:** Water molecules are associated through extensive **intermolecular hydrogen bonding**. In a water molecule, the highly electronegative oxygen atom attracts the shared electrons, creating a partial negative charge on oxygen and partial positive charges on hydrogen atoms. The hydrogen atom of one water molecule is attracted to the oxygen atom of another water molecule, forming hydrogen bonds. Considerable energy is required to break these intermolecular attractions before water can change into vapour. Therefore, water has an unusually high boiling point compared to other hydrides of Group 16.
- **Statement 2 is correct:** In ice, water molecules are arranged in a three-dimensional open cage-like structure due to hydrogen bonding. This structure contains vacant spaces, causing the molecules to be farther apart than in liquid water. As a result, ice occupies a larger volume for the same mass and is therefore less dense than liquid water. This is why ice floats on water. When ice melts, some hydrogen bonds break, allowing water molecules to come closer together and increasing the density.
- **Statement 3 is correct:** Lower alcohols such as methanol and ethanol are soluble in water because their hydroxyl (-OH) group can form hydrogen bonds with water molecules. The hydrogen

bonding between alcohol molecules and water molecules facilitates mixing and dissolution. Thus, the solubility of lower alcohols in water is largely due to hydrogen bonding. As the hydrocarbon chain length increases, the effect of the non-polar part becomes more significant and solubility decreases.

Q 56. Ans. A

Explanation:

- **Assertion (A) is true:** Water (H_2O) has an unusually high boiling point compared to other hydrides of Group 16, such as H_2S , H_2Se , and H_2Te . Normally, boiling points increase down the group due to increasing molecular mass. However, water shows an anomalously high boiling point because water molecules are strongly associated through intermolecular hydrogen bonding. A large amount of energy is required to separate these molecules before boiling can occur.
- **Reason (R) is true:** Hydrogen bonding occurs when hydrogen is bonded to highly electronegative atoms such as fluorine, oxygen, or nitrogen. In water, oxygen is highly electronegative, and each water molecule can form hydrogen bonds with neighbouring water molecules. Sulfur is much less electronegative than oxygen, and therefore H_2S does not form significant hydrogen bonding. As a result, intermolecular forces in H_2S are much weaker than those in water.
- **Why the Reason correctly explains the Assertion:**
 - ✓ The high boiling point of water is directly due to the presence of extensive hydrogen bonding between water molecules. Since H_2S lacks hydrogen bonding, its molecules are held together only by weaker intermolecular forces. Therefore, much less energy is needed to boil H_2S than water.

Q 57. Ans. B

Explanation:

- **An ionic bond** is formed by the transfer of electrons from one atom to another, resulting in the formation of oppositely charged ions. The electrostatic force of attraction between these ions acts equally in all directions around the ion. Therefore, ionic bonds are **non-directional** in nature. This non-directional character is one reason ionic compounds form crystal lattices and are generally brittle.
- **A covalent bond** is formed by the sharing of electron pairs between atoms. The shared electron pair is localised between the bonded atoms, giving

the bond a specific orientation in space. As a result, covalent bonds are **directional in nature**. The directional character of covalent bonds determines the shape and geometry of molecules, **such as the tetrahedral shape of methane and the bent shape of water**.

- In metals, valence electrons are delocalized and move freely throughout the metallic lattice, forming a “sea of electrons” around positive metal ions. Since the attraction is not confined to any particular pair of atoms, metallic bonding is **non-directional** in nature. This non-directional bonding contributes to the malleability and ductility of metals.

Q 58. Ans. B

Explanation:

- A light year is the distance travelled by light in a vacuum in one year. The speed of light is approximately 3×10^8 m/s.
- **Distance travelled in one year:**
 - ✓ Distance = Speed $\times$ Time
 - ✓ $= 3 \times 10^8 \times (365 \times 24 \times 60 \times 60)$
 - ✓ $= 3 \times 10^8 \times 31,536,000$
 - ✓ $\approx 9.46 \times 10^{15}$ m
- Thus, **one light year equals approximately 9.46×10^{15} m.**

Q 59. Ans. C

Explanation:

- **Mass of the object:**
 - ✓ $g = GM/R^2$
- **where:**
 - ✓ G = universal gravitational constant
 - ✓ M = mass of the Earth
 - ✓ R = radius of the Earth
- **The expression for g does not contain the mass of the falling object.** Therefore, the acceleration due to gravity at a place is independent of the mass of the object. Heavy and light bodies experience the same acceleration due to gravity when air resistance is neglected.

Q 60. Ans. C

Explanation:

- Fish possess a **swim bladder**, an air-filled sac that helps them control buoyancy. By changing the amount of gas in the swim bladder, a fish can alter its average density relative to water. When its density becomes lower than that of water, it rises; when its density becomes higher, it sinks. Thus, fish move up and down in water by altering their body density.

Q 61. Ans. B

Explanation:

- Soaps and detergents reduce the surface tension of water. Soap molecules accumulate at the surface and weaken the cohesive forces between water molecules. As a result, the surface tension decreases significantly. This is why soap solutions spread more easily and enhance the cleaning action of water.

Q 62. Ans. A

Explanation:

- **Statement 1 is correct:** The electromagnetic spectrum consists of radio waves, microwaves, infrared rays, visible light, ultraviolet rays, X-rays, and gamma rays. **Gamma rays occupy the extreme high-frequency end of the electromagnetic spectrum.** Since frequency and energy are directly related, gamma rays possess the highest frequency and the highest energy among all electromagnetic radiations. They also have the shortest wavelength.
- **Statement 2 is correct:** Visible light is a small portion of the electromagnetic spectrum that can be detected by the human eye. Like all electromagnetic waves, visible light consists of oscillating electric and magnetic fields that propagate through space without requiring a material medium. The visible region lies between infrared and ultraviolet radiations in the electromagnetic spectrum.
- **Statement 3 is incorrect:** Cathode rays are streams of **electrons** emitted from the cathode in a discharge tube. Since electrons are negatively charged particles possessing mass, cathode rays are particle beams and not electromagnetic waves. Their study led to the discovery of the electron by J. J. Thomson. Unlike electromagnetic waves, cathode rays are deflected by electric and magnetic fields.

Q 63. Ans. D

Explanation:

- **An echo is the repetition of a sound heard after it is reflected from a distant obstacle such as a wall, mountain, or building.** Echo involves **reflection of sound**, not the bending of sound waves around obstacles.
- **Interference occurs when two or more waves meet and superpose, resulting in reinforcement or cancellation of the resultant wave.** Interference is related to the superposition of waves and does not explain the bending of sound around barriers.

- Resonance is the phenomenon in which an object vibrates with maximum amplitude when subjected to a periodic force whose frequency matches its natural frequency. Resonance does not involve the bending of sound waves around obstacles.
- **Diffraction is the bending or spreading of waves when they pass around the edges of obstacles or through narrow openings.** Sound waves have relatively large wavelengths, so they diffract easily around barriers. Due to diffraction, sound can be heard even when the source is not in the direct line of sight, such as hearing a person speaking from behind a wall or around a corner.

Q 64. Ans. C

Explanation:

- **Statement 1 is correct:** Doppler Effect is the apparent change in frequency (or wavelength) of a wave due to the relative motion between the source of the wave and the observer. The phenomenon is observed in sound as well as electromagnetic waves. If either the source, the observer, or both are moving relative to each other, the observed frequency differs from the actual frequency emitted by the source.
- **Statement 2 is incorrect:** When the observer moves towards the source or when the source moves towards the observer, the sound waves reach the observer more frequently. **As a result, the observed frequency increases and the sound appears shriller (higher-pitched).** Conversely, when the source and observer move away from each other, the observed frequency decreases.
- **Statement 3 is correct:** The use of ultrasonic waves and the Doppler principle in detecting moving objects under water. SONAR (Sound Navigation and Ranging), used in ships and submarines, employs ultrasonic waves to detect underwater objects and determine their position and movement. The Doppler Effect helps in measuring the speed and motion of underwater targets, making it useful in submarine navigation and detection systems.

Q 65. 65. Ans. C

Explanation:

- Parkinson's disease is associated with a **decrease in the production of dopamine** in the brain. Dopamine acts as a neurotransmitter and plays an important role in the control of body movements. Deficiency of dopamine results in symptoms such as tremors, muscular rigidity, and impaired motor coordination, which are characteristic features of Parkinson's disease.

Q 66. Ans. B

Explanation:

- **Bile juice is produced by the liver and stored in the gall bladder,** is released into the **duodenum**, which is the first part of the small intestine. Bile does not contain digestive enzymes but helps in the digestion of fats by emulsifying them into smaller droplets. It also provides an alkaline medium that facilitates the action of pancreatic enzymes.
- The stomach is a J-shaped organ where food is temporarily stored and mixed with gastric juice containing hydrochloric acid, mucus, and digestive enzymes.
- The ileum is the terminal part of the small intestine and is mainly involved in the absorption of digested nutrients. Bile is released earlier, into the duodenum, not the ileum.
- The colon is a part of the large intestine. Its primary functions include absorption of water and salts and the formation of faeces. Bile is not released into the colon.

Q 67. Ans. B

Explanation:

- Pepsin is a protein-digesting enzyme secreted in the stomach. It converts proteins into smaller peptides. Pepsin is present in both infants and adults and is not specifically responsible for coagulating milk.
- In **infants**, the stomach secretes **rennin (chymosin)**, an enzyme that coagulates or curdles milk protein (casein). This slows the passage of milk through the alimentary canal and allows sufficient time for digestion. Rennin is particularly important during infancy when milk is the primary source of nutrition.
- Maltase is an intestinal enzyme that converts maltose into glucose. It is involved in carbohydrate digestion and has no role in coagulating milk.
- Amylase is an enzyme that digests starch into simpler sugars. It is present in saliva and pancreatic secretions and is not involved in the coagulation of milk.

Q 68. Ans. C

Explanation:

- **Statement 1 is correct:** Optical fibres transmit light signals through the phenomenon of **total internal reflection**. When light enters the fibre at a suitable angle, it undergoes repeated total internal reflections at the boundary between the core and the cladding,

allowing the light to travel long distances with very little loss of energy. This principle is the basis of the operation of optical fibres.

- **Statement 2 is incorrect:** For total internal reflection to occur, light must travel from a denser medium to a rarer medium. Therefore, in an optical fibre, the **core has a higher refractive index** than the cladding. The cladding is made of a material with a slightly lower refractive index so that light remains confined within the core through repeated total internal reflection.
- **Statement 3 is correct:** Optical fibres are extensively used in **telecommunications** because they can transmit large amounts of information rapidly and with minimal signal loss. They are used in telephone networks, internet communication, cable television systems, and other communication technologies. Optical fibres are also used in medical instruments such as endoscopes.

Q 69. Ans. C

Explanation:

- Phage therapy is the use of **bacteriophages (phages)**, viruses that specifically infect and destroy bacteria, to treat bacterial infections. After infecting a bacterial cell, the phage multiplies within it and eventually causes the bacterium to lyse (burst), thereby eliminating the infection-causing bacteria.

Q 70. Ans. D

Explanation:

- **Statement 1 is correct:** Vaccines stimulate the immune system to produce antibodies and memory cells against specific disease-causing agents. The **DPT (or DTP) vaccine** is a combined vaccine that provides protection against **diphtheria, pertussis (whooping cough), and tetanus**. It is one of the commonly administered childhood vaccines used in immunisation programmes.
- **Statement 2 is correct:** All three diseases targeted by the DTP vaccine are caused exclusively by bacterial pathogens. Diphtheria is caused by the bacterium *Corynebacterium diphtheriae*, which produces a toxin affecting the throat and respiratory system. Pertussis is caused by *Bordetella pertussis*, a bacterium that infects the respiratory tract and causes severe coughing fits. Tetanus is caused by *Clostridium tetani*, a bacterium found in soil that produces a neurotoxin affecting muscle function.

Q 71. Ans. C

Explanation:

- The **sinoatrial node (SA node)** is located in the wall of the right atrium near the opening of the superior vena cava. It possesses the highest rate of spontaneous electrical impulse generation and initiates each heartbeat. Because it determines the rhythm and rate of cardiac contractions, the SA node is known as the **natural pacemaker of the heart**.
- Bundle of His is a specialised bundle of conducting fibres that carries impulses from the AV node to the ventricles. It forms part of the cardiac conduction system but does not act as the natural pacemaker.
- Purkinje fibres are specialised conducting fibres distributed throughout the ventricular walls. Their function is to rapidly transmit electrical impulses to the ventricular muscle, ensuring coordinated ventricular contraction. They do not initiate the heartbeat under normal conditions.
- **Atrioventricular node (AV node)** is located in the lower right atrium near the interatrial septum. It receives the electrical impulse generated by the SA node and relays it to the ventricles. Although it can generate impulses under certain conditions, it is not the natural pacemaker of the heart.

Q 72. Ans. D

Explanation:

- The human skeletal system is divided into two main parts: the **axial skeleton** and the **appendicular skeleton**. The **axial skeleton** forms the central axis of the body and consists of the **skull, vertebral column, ribs, and sternum**. The skull protects the brain and supports the structures of the face, while the vertebral column protects the spinal cord and provides support to the body. Since the question mentions the **skull and vertebral column**, the correct answer is the **axial skeleton**.

Q 73. Ans. C

Explanation:

- **Statement 1 is correct:** Cardiac muscles are found exclusively in the walls of the heart. These muscles function without conscious control and are therefore **involuntary**. Cardiac muscle fibers are **cylindrical, branched, and interconnected**, forming a network that enables coordinated contraction of the heart. They also show alternating light and dark bands, making them **striated**. The rhythmic contraction

and relaxation of cardiac muscles help pump blood throughout the body.

- **Statement 2 is incorrect:** Skeletal muscles are muscles attached to bones and responsible for voluntary movements such as walking, running, and lifting objects. These muscles are **striated, voluntary, long, cylindrical, and multinucleate**. They are not spindle-shaped. The **spindle-shaped** muscle cells are characteristic of **smooth muscles**, which occur in the walls of internal organs such as the digestive tract, urinary bladder, and blood vessels. Therefore, the description given in the statement actually applies to smooth muscles, not skeletal muscles.
- **Statement 3 is correct:** Smooth muscles are present in the walls of visceral organs such as the stomach, intestine, urinary bladder, blood vessels, and uterus. These muscles do not exhibit the alternating dark and light bands seen in skeletal and cardiac muscles and are therefore **non-striated**. Smooth muscle cells are **spindle-shaped** and contain a **single centrally located nucleus**, making them **uninucleate**. Their actions are involuntary and controlled by the autonomic nervous system.

Q 74. Ans. C

Explanation:

- **Turner's Syndrome** is a chromosomal disorder caused by the absence of one sex chromosome, resulting in a total of **45 chromosomes (45, X)** instead of the normal 46. This condition is an example of **aneuploidy**, specifically **monosomy of the X chromosome**. Individuals with Turner's Syndrome are phenotypically female but often exhibit underdeveloped ovaries, a short stature, a broad chest, and other characteristic features. They are usually sterile due to impaired reproductive development.

Q 75. Ans. D

Explanation:

- **Pair 1 is correct:** **Gregor Johann Mendel** is regarded as the "Father of Genetics." Through experiments on garden pea plants (*Pisum sativum*), he formulated the fundamental principles of inheritance. His work led to the establishment of the **Law of Dominance**, **Law of Segregation**, and the basis for the **Law of Independent Assortment**. Mendel demonstrated that traits are inherited through discrete factors, now known as genes.
- **Pair 2 is correct:** **James Watson and Francis Crick** proposed the **double helix model of DNA** in 1953.

Their model explained how genetic information is stored and replicated. The structure consists of two polynucleotide chains coiled around a common axis, with complementary base pairing between adenine-thymine and guanine-cytosine. Therefore, associating James Watson with the discovery of the DNA double helix structure is correct.

- **Pair 3 is correct:** **Frederick Griffith** conducted experiments in 1928 using two strains of *Streptococcus pneumoniae* bacteria. He observed that non-virulent bacteria could be transformed into virulent forms when mixed with heat-killed virulent bacteria. This phenomenon was termed **transformation** and provided the first evidence that a "transforming principle" could transfer hereditary information from one cell to another.
- **Pair 4 is correct:** **Rosalind Franklin** produced crucial **X-ray diffraction photographs of DNA**, particularly the famous "Photo 51." These images provided key evidence regarding the helical structure of DNA and helped Watson and Crick develop their model. Her work revealed important details about the dimensions and arrangement of DNA molecules.

Q 76. Ans. D

Explanation:

- **The Karkhanawala Committee** was constituted to examine existing nuclear safety policies and suggest measures to strengthen India's nuclear regulatory framework. A major recommendation of the committee was that nuclear safety oversight should not remain solely within departmental mechanisms. The committee was initially chaired by **Dr. S.D. Soman** and later by **Dr. M.D. Karkhanawala**. After Dr. Karkhanawala's untimely demise in 1980, **Shri V.N. Meckoni** assumed leadership of the committee.
- **The Meckoni Committee** was constituted on **21 March 1987** to review the powers, functions, and responsibilities of the **Atomic Energy Regulatory Board (AERB)**, which had recently been established. The committee's recommendations were accepted almost entirely by the Government of India. These recommendations significantly expanded the regulatory authority of the AERB, incorporated the Department of Atomic Energy's Safety Review Committee into the AERB framework, and strengthened the institutional structure for nuclear safety regulation.
- The term **Kakodkar Committee** commonly refers to two prominent committees chaired by renowned nuclear scientist **Dr. Anil Kakodkar**. One was the

Committee on Autonomy and Future of IITs (2011), which examined reforms and greater autonomy for the Indian Institutes of Technology. The other was the **High-Level Safety Review Committee (2012)**, constituted to review and recommend measures for enhancing safety in **Indian Railways**.

Q 77. Ans. D

Explanation:

- **Statement 1 is correct:** Mission Drishti is a cutting-edge, dual-use Earth observation satellite and is uniquely recognized as the world's first **OptoSAR satellite**. The satellite combines **optical imaging sensors** and **Synthetic Aperture Radar (SAR) sensors** on a single operational platform. This integration enables the generation of high-clarity, analysis-ready Earth observation data by leveraging the strengths of both technologies.
- **Statement 2 is correct:** Mission Drishti was developed by **GalaxEye**, a Bengaluru-based spacetechnology startup specializing in advanced Earth observation solutions. The mission received support from **IN-SPACe (Indian National Space Promotion and Authorization Center)**, which facilitates and promotes private sector participation in India's space ecosystem.
- **Statement 3 is correct:** Mission Drishti was successfully launched aboard a **SpaceX Falcon 9** rocket from **Vandenberg, California, USA**. The satellite was placed in a **Sun-synchronous Low Earth Orbit (LEO)** at an altitude of approximately **500 km**. The mission aims to provide **all-weather, day-and-night Earth observation capabilities** by combining optical and radar data.

Q 78. Ans. C

Explanation:

- **Cyborg Botany** is an emerging interdisciplinary field that combines **biology, engineering, and materials science** to integrate living plants with electronic components. It aims to bridge the gap between biological signals and electronic data, enabling plants to communicate their internal health in real time.
- **Key Features:**
 - ✓ **Direct Embedding:** Nanowires and electronic transistors are embedded into plant cell walls to function as biosensors.
 - ✓ **Conductive Polymers:** Materials such as **PEDOT** act as living electrical pathways within plant tissues.

- ✓ **Signal Conversion:** Plants' natural responses to light, moisture, pests, and other environmental factors are converted into digital data.

- **Aim:** To detect **biotic and abiotic stresses** at the cellular or genetic level, allowing intervention before visible damage occurs.

Q 79. Ans. C

Explanation:

- **Hantavirus** is a group of viruses that can cause severe and potentially fatal **respiratory or kidney-related diseases** in humans. It is mainly associated with **Hantavirus Pulmonary Syndrome (HPS)** and **Hemorrhagic Fever with Renal Syndrome (HFRS)**.
- The virus is named after the **Hantan River** region of **South Korea**, where it was first identified in the 1970s.
- **Vector:** Rodents such as **deer mice, cotton rats, and rice rats** are the primary carriers.
- **Transmission:**
 - ✓ **Most common:** Inhalation of virus particles from contaminated rodent urine, droppings, or saliva.
 - ✓ **Direct contact** with contaminated materials.
 - ✓ **Rarely** through rodent bites.
 - ✓ **Human-to-human transmission** is extremely rare and has been reported mainly with the **Andes virus** strain in South America.

Q 80. Ans. A

Explanation:

- **SMILE (Solar wind Magnetosphere Ionosphere Link Explorer)** is a joint space mission of the **European Space Agency (ESA)** and the **Chinese Academy of Sciences (CAS)**. It is designed to study the interaction between Earth's **magnetosphere** and the **solar wind** using multi-wavelength observations.
- **Aim:**
 - ✓ To capture the first global **X-ray and ultraviolet images** of Earth's magnetic shield.
 - ✓ To understand how the magnetosphere responds to **solar flares** and **Coronal Mass Ejections (CMEs)**.
 - ✓ To improve knowledge of **solar-terrestrial interactions** and space weather processes that affect Earth.

Q 81. Ans. C

Explanation:

- According to the **Food and Agriculture Organization (FAO)**'s **Global Forest Resources Assessment**

(GFRA) 2025, India has moved up to the 9th position globally in terms of total forest area. India also continues to hold the 3rd position worldwide in net annual forest area gain, reflecting sustained progress in afforestation and forest conservation efforts.

- The report further notes that India ranks 5th among the world's top carbon sinks, with its forests removing approximately 150 million tonnes (Mt) of CO₂ annually during 2021–2025.
- At the global level, forests cover about 4.14 billion hectares, accounting for nearly 32% of the Earth's land area. The assessment also highlights a positive trend in forest conservation, as the annual rate of net forest loss declined from 10.7 million hectares during 1990–2000 to 4.12 million hectares during 2015–2025.

Q 82. Ans. C

Explanation:

- **Statement 1 is correct:** India delivered its best-ever performance at the 11th Asian Aquatics Championship, finishing 11th in the overall medal standings with a haul of 13 medals, including 4 Silver and 9 Bronze medals. The medals came across various disciplines, reflecting the growing strength of Indian aquatic sports at the continental level.
- **Statement 2 is correct:** China emerged as the overall champion by finishing at the top of the medal tally. The 11th Asian Aquatics Championship was also noteworthy because it was hosted in India for the first time, providing Indian athletes with valuable international exposure on home soil and highlighting India's increasing role in hosting major sporting events.

Q 83. Ans. D

Explanation:

- At the 70th Filmfare Awards 2025, the Lifetime Achievement Award was conferred on veteran actress Zeenat Aman and acclaimed filmmaker Shyam Benegal (posthumously) for their outstanding contributions to Indian cinema.
- **Additional information:**
 - ✓ Achint Thakkar received the R.D. Burman Award for Upcoming Talent in Music.
 - ✓ Laapataa Ladies, directed by Kiran Rao, won the Best Film award at the 70th Filmfare Awards 2025. The film also emerged as the biggest winner of the evening, securing multiple awards, including Best Director for Kiran Rao.

Q 84. Ans. D

Explanation:

- 'Bold Kurukshetra 2025' is the latest edition of the bilateral military exercise between the armies of India and Singapore. The exercise is conducted to enhance interoperability, joint training, and operational cooperation between the two forces.
- **The exercise focuses on areas such as:**
 - ✓ Joint planning and execution of military operations
 - ✓ Sharing of best practices and tactical expertise
 - ✓ Strengthening defence cooperation and mutual understanding
- The exercise is held under the framework of the India–Singapore Defence Cooperation Agreement, reflecting the close strategic partnership between the two countries.

Q 85. Ans. D

Explanation:

- The Assam Government launched 'Ankita', an AI-powered virtual news anchor capable of delivering government updates in Assamese, making it one of India's first AI anchors designed for official communication in a regional language. The initiative was unveiled by Chief Minister Himanta Biswa Sarma to communicate cabinet decisions and government announcements more effectively to citizens.

Q 86. Ans. D

Explanation:

- India's first Agri-Drone Indigenisation Facility was inaugurated by Union Minister of State for Rural Development Kamlesh Paswan at Chennai, Tamil Nadu. Established by Garuda Aerospace, the facility is India's first dedicated hub for the design, manufacturing, and testing of advanced agricultural drones and unmanned aerial systems (UAS). Spread across 35,000 sq. ft., it aims to strengthen indigenous drone manufacturing under the Atmanirbhar Bharat initiative and support the government's vision of making India a global drone hub.

Q 87. Ans. B

Explanation:

- INS Arnala is the first indigenously designed and built Anti-Submarine Warfare Shallow Water Craft (ASW-SWC) commissioned into the Indian Navy. It was commissioned at the Naval Dockyard, Visakhapatnam. Designed for operations in shallow

coastal waters, the vessel is equipped for subsurface surveillance, anti-submarine operations, search and rescue missions, and low-intensity maritime operations. It is the lead ship of a new class of ASW-SWCs being inducted to strengthen India's coastal defence capabilities.

Q 88. Ans. B

Explanation:

- World Hunger Day is observed on 28th May annually to raise awareness about sustainable solutions to hunger and poverty.
- International Everest Day is observed on 29th May, commemorating the first successful summit of Mount Everest by Sir Edmund Hillary and Tenzing Norgay in 1953.
- World Milk Day is observed on 1st June, established by FAO to recognise the importance of milk as a global food.
- World Food Safety Day is observed on 7th June, established by the United Nations to draw attention to foodborne risks and promote food safety.

Q 89. Ans. B

Explanation:

- India's first 3-nanometre (3nm) chip design centres were inaugurated in Noida (Uttar Pradesh) and Bengaluru (Karnataka) by Union Electronics and IT Minister Ashwini Vaishnaw. These centres were established by Renesas Electronics India and mark India's entry into next-generation semiconductor design technology. The 3nm node is among the most advanced chip technologies in the world, offering higher performance and improved energy efficiency.

Q 90. Ans. D

Explanation:

- **Statement 1 is correct:** India won the inaugural Blind Women's T20 World Cup by defeating Nepal by 7 wickets in the final held at the P. Sara Oval, Colombo, Sri Lanka. The Indian team remained unbeaten throughout the tournament, creating history by becoming the first-ever champions of the event.
- **Statement 2 is correct:** The Indian women's kabaddi team won its second consecutive Women's Kabaddi World Cup title by defeating Chinese Taipei in the final held in Dhaka, Bangladesh. India remained unbeaten during the tournament and successfully defended its World Cup crown.

Q 91. Ans. C

Explanation:

- Samrat Rana created history by becoming the first Indian shooter to win an individual men's 10m Air Pistol world title at the ISSF World Championship held in Cairo, Egypt. The 20-year-old shooter from Haryana won the gold medal with a score of 243.7, defeating China's Hu Kai in a closely contested final. His achievement marked India's first-ever individual world championship gold in the men's 10m air pistol event.

Q 92. Ans. D

Explanation:

- **Khanjar-XII → India-Kyrgyzstan**
 - ✓ Khanjar-XII was the 12th edition of the India-Kyrgyzstan Joint Special Forces Exercise, aimed at enhancing cooperation in counter-terrorism and special operations.
- **Dustlik → India-Uzbekistan**
 - ✓ Dustlik is a bilateral military exercise conducted between the armed forces of India and Uzbekistan to strengthen defence cooperation and interoperability.
- **Varuna-23rd edition → India-France**
 - ✓ Varuna is the flagship bilateral naval exercise between the Indian Navy and the French Navy. The 23rd edition of Varuna was conducted in 2025.
- **Nomadic Elephant → India-Mongolia**
 - ✓ Nomadic Elephant is an annual military exercise conducted between the armies of India and Mongolia, focusing on counter-insurgency and peacekeeping operations.

Q 93. Ans. C

Explanation:

- The 17th GRIHA Summit, organised by the GRIHA Council in association with TERI (The Energy and Resources Institute), was held under the theme "Innovate to Act for a Climate Resilient World." The summit brought together policymakers, industry leaders, architects, urban planners, researchers, and sustainability experts to discuss innovative and actionable solutions for building climate-resilient infrastructure and sustainable habitats. The discussions focused on climate adaptation, resilient cities, circular economy, energy efficiency, and sustainable development.

Q 94. Ans. A**Explanation:**

- **Statement 1 is correct:** The first-ever Conference for Women Peacekeepers was organised in New Delhi. The central theme of this historic event was specifically dedicated to amplifying the Global South Perspective. It aimed to explore the challenges, unique field insights, and operational contributions of women peacekeepers from developing nations, who form the vast majority of troop-contributing countries.
- **Statement 2 is incorrect:** The conference was not organised under the direct aegis of the United Nations Security Council (UNSC). Instead, the conference was organised by the Ministry of External Affairs (MEA) in collaboration with the Ministry of Defence (MoD), and hosted by the Indian Army through the Centre for United Nations Peacekeeping (CUNPK).

Q 95. Ans. D**Explanation:**

- The first-ever Regional Dialogue on Social Justice under the Global Coalition for Social Justice was hosted by India at Bharat Mandapam, New Delhi. The event was organised by the Ministry of Labour & Employment, Government of India, in collaboration with the Confederation of Indian Industry (CII)–Employers’ Federation of India (EFI).
- As a member of the Asia-Pacific Coordinating Group, India spearheaded this first Regional Dialogue under the Global Coalition for Social Justice, an initiative launched by the International Labour Organisation (ILO) in 2023.

Q 96. Ans. D**Explanation:**

- **Statement 1 is correct:** The National Commission for Women (NCW) launched the “SHAKTI Scholars: NCW Young Research Fellowship” to promote policy-oriented research on women-centric issues. It is a six-month, grant-based research fellowship, and selected candidates receive a research grant of ₹1 lakh.
- **Statement 2 is correct:** The fellowship is open to Indian citizens aged 21–30 years who possess at least a graduation degree from a recognised institution. Preference is given to candidates pursuing or having completed postgraduate or higher research qualifications.

Q 97. Ans. C**Explanation:**

- **Statement 1 is correct:** The Bihar Government launched ‘Bhumi Sudhar Samvad’ to accelerate the resolution of land disputes and improve transparency in land administration. The programme was launched with its first session in Patna, and subsequent sessions were planned in Lakhisarai and other districts to facilitate direct interaction between citizens and land administration officials.
- **Statement 2 is correct:** As part of the initiative, the government announced the formation of a specialised monitoring team called ‘Udan Dasta’. The team has been tasked with detecting, monitoring, and preventing land-related fraud and irregularities, thereby strengthening transparency and accountability in land governance.

Q 98. Ans. B**Explanation:**

- The Global SPECS 2030 Initiative was launched by the World Health Organisation (WHO) to ensure that everyone who needs refractive error interventions (such as spectacles) has access to quality, affordable, and people-centred eye-care services by 2030. The initiative seeks to address uncorrected refractive error, which is the leading cause of vision impairment globally.
- **The initiative supports the global target adopted at the 74th World Health Assembly (2021) of increasing effective refractive error coverage by 40 percentage points by 2030. The acronym SPECS stands for:**
 - ✓ Services
 - ✓ Personnel
 - ✓ Education
 - ✓ Cost
 - ✓ Surveillance

Q 99. Ans. B**Explanation:**

- India’s first Power Museum is being developed by the Bihar State Power Holding Company Limited (BSPHCL) at the decommissioned Karbigahiya Thermal Power Station in Patna, Bihar. The museum aims to showcase the history and evolution of the power sector in Bihar and India, including old power-generation equipment, transmission technologies, and milestones in electrification.

Q 100. Ans. C

Explanation:

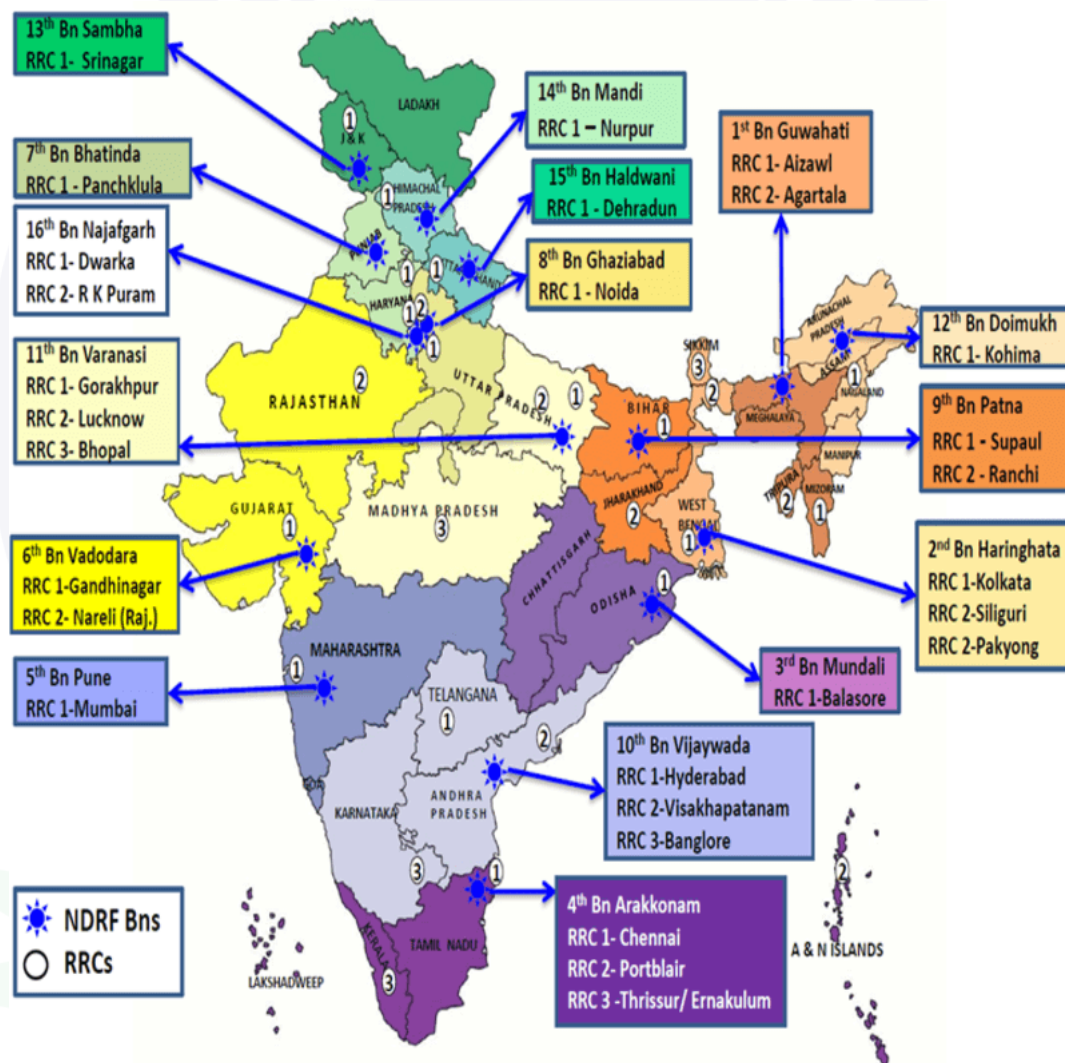
- Ireland was named one of the Country Partners for the Hornbill Festival 2025, held from 1 to 10 December 2025 at the Kisama Heritage Village near Kohima. The other three options were not partner countries for this edition.
- The 2025 edition had six Country Partners in total: the United Kingdom, Ireland, France, Switzerland, Austria, and Malta. Arunachal Pradesh joined as the State Partner.
- The Hornbill Festival, called the “Festival of Festivals”, showcases the traditions, crafts, music,

dance, and cuisine of Nagaland’s Naga tribes. It is named after the Great Indian Hornbill and was first held in 2000.

Q 101. Ans. C

Explanation:

- The 9th Battalion of the National Disaster Response Force (NDRF) is headquartered at Bihta, Patna, Bihar, and is responsible for disaster response operations across Bihar and Jharkhand. It serves as a specialised unit for managing floods and other water-related disasters, particularly during the monsoon season.



● Key Details

- ✓ **Headquarters:** Bihta, Patna, Bihar
- ✓ **Area of Responsibility:** Bihar and Jharkhand
- ✓ **Specialisation:** Flood rescue and water-related disaster management
- ✓ **Regional Response Centre (RRC):** Supaul, Bihar

● Major Functions

- ✓ Deploying rescue teams in advance during

the monsoon season and carrying out search, rescue, and relief operations during floods, earthquakes, cyclones, and other disasters.

- ✓ Conducting disaster preparedness activities, including earthquake safety drills, first-responder medical training, and awareness programmes for schools, NCC cadets, and local communities.
- ✓ Providing specialised disaster management

training to personnel of other security agencies, including CISF units, to strengthen emergency response capabilities.

Q 102. Ans. D

Explanation:

- According to the Census 2011, Purba Champaran (East Champaran) has the highest rural population among all districts in Bihar.
- The top five districts in Bihar by rural population are:
 - ✓ Purba Champaran (East Champaran)
 - ✓ Muzaffarpur
 - ✓ Madhubani
 - ✓ Samastipur
 - ✓ Gaya
- These figures, based on the 2011 Census, highlight the predominance of rural populations in these districts, with Purba Champaran leading the state

Q 103. Ans. C

Explanation:

- According to the Census 2011, Bihar is the most densely populated state in India, with an average population density of 1,106 persons per square kilometre.
- The five districts with the highest population density in Bihar are:
 1. Sheohar
 2. Patna
 3. Darbhanga
 4. Vaishali
 5. Begusarai
- Although Patna is the most populous district in Bihar in terms of total population, Sheohar ranks first in population density because its relatively small geographical area accommodates a large number of people.

Q 104. Ans. B

Explanation:

- The Kalrav Festival is Bihar's flagship state-level bird festival, organised annually by the state's Environment, Forest and Climate Change Department. The festival aims to promote eco-tourism, encourage bird conservation, and create awareness about migratory bird species.
- The festival is held at the Nagi-Nakti Bird Sanctuaries in Jamui district, located about 200 km southeast of Patna.
- The twin reservoirs and adjoining wetlands of Nagi

and Nakti attract thousands of migratory birds during the winter season, including the **endangered Bar-headed Goose**, making the area an important bird habitat.

- The first edition of the festival was inaugurated by Nitish Kumar in January 2021.

Q 105. Ans. C

Explanation:

- The 37th Session of the Indian National Congress was held at Gaya, Bihar, in December 1922 under the presidency of Chittaranjan Das.
- The session witnessed a significant ideological divide within the Congress between the “No-Changers” and the “Pro-Changers.” The No-Changers supported Mahatma Gandhi's strategy of continuing the boycott of legislative councils and emphasising constructive programmes, while the Pro-Changers favoured entering the councils to challenge British rule from within.

Q 106. Ans. B

Explanation:

- **Statement 1 is incorrect:** Jairam Das Daulatram was the first Governor of Bihar after Independence, serving from 1947 to 1948. Sir Charles Stuart Bayley was the first Lieutenant Governor of Bihar and Orissa after the creation of the province in 1912, when Bihar and Orissa were separated from the Bengal Presidency.
- **Statement 2 is correct:** One of the earliest decisions taken in his name was the construction of a memorial for the seven students martyred on 11 August 1942 while attempting to unfurl the national flag at the Patna (Old) Secretariat building during the Quit India Movement.

Q 107. Ans. D

Explanation:

- **Statement 1 is correct:** During British rule, the vast open ground in the heart of Patna was known as **Race Course Maidan** (also called Patna Lawns in some records). It was primarily used for horse racing, parades, and official colonial gatherings. After Independence, the ground was renamed **Gandhi Maidan** in honour of Mahatma Gandhi and his association with the freedom struggle.
- **Statement 2 is correct:** In the aftermath of communal tensions associated with the Partition of India (1947), Mahatma Gandhi visited Bihar and undertook efforts to promote peace and communal

harmony. He observed a fast at one corner of Gandhi Maidan and appealed to people to end violence and maintain unity among communities.

- **Statement 3 is correct:** A statue of Mahatma Gandhi has been installed at Gandhi Maidan to commemorate his association with the site and his efforts to restore peace during a turbulent period in Indian history. The statue stands near the location linked to his fast and serves as a reminder of his message of non-violence and communal harmony.

Q 108. Ans. A

Explanation:

- **Assertion (A) is true:** Valmiki Nagar, located in West Champaran district of Bihar near the India–Nepal border, is closely associated with the **Gandak River (Narayani River in Nepal)**. The famous **Valmiki Nagar Barrage** has been constructed across the Gandak River and serves as an important source of irrigation for Bihar, Uttar Pradesh, and parts of Nepal.
- **Reason (R) is true:** The Gandak River originates in the Himalayas of Nepal and enters Bihar at **Valmiki Nagar**. After entering the state, it flows southeastward through the plains before ultimately joining the Ganga River near Patna.
- **Why is (R) the correct explanation of (A)?**
 - ✓ Valmiki Nagar's association with the Gandak River exists primarily because the river **enters Bihar at this location**, making it a significant geographical and hydrological point. The construction of the Valmiki Nagar Barrage and the development of the Gandak irrigation system are direct consequences of the river's entry into Bihar at Valmiki Nagar. Therefore, the reason directly explains why Valmiki Nagar is closely associated with the Gandak River.

Q 109. Ans. C

Explanation:

- **Assertion (A) is true:** Bihar has a **humid subtropical climate**, characterised by hot summers, a distinct southwest monsoon season that brings most of the annual rainfall, and cool winters. The state's climate is influenced by its location in the middle Gangetic plain and the seasonal monsoon winds.
- **Reason (R) is false:** Bihar does **not** lie south of the Tropic of Cancer. Instead, it is situated entirely **north of the Tropic of Cancer**, extending roughly from $24^{\circ}20' \text{ N}$ to $27^{\circ}31' \text{ N}$ latitude. Therefore, the reason contains a geographical error. The **Tropic**

of Cancer ($23^{\circ}30' \text{ N}$) passes through **eight Indian states**—Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, and Mizoram. Bihar is **not** one of these states.

Q 110. Ans. D

Explanation:

- **Statement 1 is correct:** One of its most celebrated saints was Hazrat Sharfuddin Ahmad. He was born at Maner in 1262 AD and died at Bihar Sharif in 1377 AD." Thus, Hazrat Sharfuddin Ahmad Yahya Maneri was indeed born at **Maner**, which is one of the oldest centres of Sufism in Bihar.
- **Statement 2 is correct:** "The most celebrated of all the Sufi orders in Bihar was the Suhrawardia, particularly its sub-section, the Firdausia Silsila." Therefore, the **Firdausia Silsila** was a **sub-section (sub-branch) of the Suhrawardia order** and played a significant role in the spread of Sufism in Bihar.
- **Statement 3 is correct:** Peer Mujeebullah was a great Persian and Arabic writer and has also laid the foundation stone of a Madrasa and library named Dar-ul-Oloom Mujeebiya." The institution is notable for its collection of around **10,000 books**, rare **Arabic and Persian manuscripts**, and a **handwritten copy of the Holy Quran associated with Mughal Emperor Aurangzeb**.

Q 111. Ans. C

Explanation:

- **Paika Rebellion (1817):** Led by Bakshi Jagabandhu (the military chief of the Raja of Khurda) in Odisha against the British East India Company's oppressive land revenue policies and currency system.
- **Ahom Revolt (1828):** Led by Gomidhar Konwar in Assam. After the First Anglo-Burmese War, instead of withdrawing from Assam as promised, the British attempted to incorporate the Ahom territories into the Company's dominion.
- **Kol Mutiny (1831–32):** Led by Buddho Bhagat (and later others like Joa Bhagat) in the Chhotanagpur region. The rebellion was sparked by the large-scale transfer of tribal lands to outsiders like Sikh and Muslim farmers.
- **Pagla Panthis (1825–1835):** Led by Tipu Shah (son of the semi-religious sect founder, Karam Shah) in the Mymensingh region of Bengal. It was a peasant movement aimed at defending the rights of tenant farmers against the exorbitant rents demanded by local zamindars.

Q 112. Ans. A

Explanation:

- The commercialisation of agriculture was a major feature of British economic policy in India during the 19th century. Under this process, agriculture became increasingly oriented toward the market rather than local consumption.
- Farmers were encouraged or compelled to cultivate cash crops such as cotton, indigo, jute, tea, coffee, and opium instead of traditional food grains. The British demand for raw materials for industries in Britain and export markets drove this transformation.

Q 113. Ans. C

Explanation:

- The Tattwabodhini Sabha was founded by Debendranath Tagore in 1839. Its objective was to promote the teachings of the Upanishads and spread the ideals of the Brahmo movement. It helped revive and strengthen the Brahmo Samaj, which had become inactive after the departure and death of Raja Ram Mohan Roy.
- The Young Bengal Movement was led by Henry Vivian Derozio, a teacher at Hindu College, Calcutta. His followers, known as the Derozians, advocated rational thinking, freedom of expression, women's rights, and the rejection of social and religious orthodoxy. The movement played an important role in stimulating intellectual awakening in Bengal.
- The Dev Samaj was founded in 1887 by Shiv Narayan Agnihotri at Lahore. It emphasised ethical living, social reform, women's upliftment, and moral values. The movement sought to reform society through character-building rather than merely religious reinterpretation.

Q 114. Ans. C

Explanation:

- **Statement 1 is correct:** The Policy of Ring-Fence (1765–1813) was followed by the British to protect their possessions by creating a belt of allied states around Company territories. These states acted as buffer zones against external threats such as the Marathas, Afghans, or other powers. The cost of maintaining this defensive arrangement was largely borne by the allied Indian states rather than the Company.
- **Statement 2 is correct:** Under the Policy of Subordinate Isolation (1813–1857), Indian states

were required to accept British supremacy. They lost the right to conduct independent foreign relations and could not wage war or enter into treaties without British approval. While internal administration generally remained with the rulers, their external sovereignty was effectively surrendered to the British.

Q 115. Ans. D

Explanation:

- **Statement 1 is correct:** In the Mughal period, society was organized around local, caste, clan, tribal, regional, and religious identities. The modern concept of loyalty to a nation-state had not yet developed.
- **Statement 2 is correct:** When Mughal authority weakened, various groups such as the Marathas, Jats, Sikhs, Rajputs, and regional governors sought greater autonomy or independence rather than preserving the empire's unity.
- **Statement 3 is correct:** The decline of Mughal power saw numerous zamindars and local chiefs asserting independence, refusing revenue payments, and establishing their own authority, indicating the absence of a larger national political commitment.

Q 116. Ans. B

Explanation:

- **Recovery of Lohgarh (1711):** Banda Bahadur and Sikh forces reoccupied Lohgarh after Mughal operations.
- **Defeat of Abdullah Khan (1720):** The powerful Sayyid Brothers' dominance ended when Wazir Abdullah Khan was defeated near Agra.
- **Battle of Karnal (1739):** Battle of Karnal resulted in a crushing defeat of the Mughal Empire by Nader Shah.
- **Battle of Buxar (1764):** Battle of Buxar established the political supremacy of the East India Company in eastern India.

Q 117. Ans. D

Explanation:

- Shah Abdul Latif Bhittai (1689–1752), the greatest Sindhi Sufi poet, composed the celebrated poetic work "Risalo" (Shah Jo Risalo) in the 18th century.
- **The Risalo is a collection of Sufi poetry that:**
 - ✓ Emphasizes love, devotion, and spiritual union with God.
 - ✓ Draws upon Sindhi folk tales and local traditions.
 - ✓ It is regarded as the masterpiece of Sindhi literature.

Q 118. Ans. A

Explanation:

- **Statement 1 is correct:** Tipu Sultan introduced several administrative and economic reforms aimed at modernising the Mysore state. These included:
 - ✓ A new calendar (Mauludi era) replacing the traditional Hijri calendar.
 - ✓ A revised coinage system with new coins and denominations.
 - ✓ Standardised weights and measures to improve administration and commerce.
- **Statement 2 is incorrect:** Tipu Sultan did not favour decentralised trade administration. Instead, he sought to establish state control over trade and commerce. He created state trading depots, encouraged government monopolies in key commodities such as sandalwood, pepper, cardamom, and betel nut, and attempted to reduce the influence of private intermediaries.

Q 119. Ans. B

Explanation:

- In 1498, the Portuguese navigator Vasco da Gama became the first European to reach India by sea. He sailed from Portugal, rounded the Cape of Good Hope at the southern tip of Africa, crossed the Indian Ocean with the help of an Arab pilot, and landed at Kozhikode (Calicut) on the Malabar Coast.
- This voyage established a direct sea route between Europe and India, breaking the monopoly of Arab and Venetian traders over the spice trade.

Q 120. Ans. B

Explanation:

- The Battle of Wandiwash (1760) was a decisive battle of the Third Carnatic War fought between the British and the French in South India.
- The French forces were commanded by Thomas Arthur de Lally, while the British forces were led by Eyre Coote.
- Eyre Coote decisively defeated Lally at Wandiwash (near present-day Vandavasi in Tamil Nadu) in January 1760. This victory effectively ended French military ambitions in India and paved the way for British supremacy.

Q 121. Ans. C

Explanation:

- After the withdrawal of the Non-Cooperation Movement in 1922, the Congress split into two groups:

- ✓ **Swarajists (Pro-Changers)** – Led by Chittaranjan Das and Motilal Nehru, who advocated entering legislative councils to obstruct colonial rule from within.
- ✓ **No-Changers** – Led by leaders such as Vallabhbhai Patel, Rajendra Prasad, C. Rajagopalachari, and others. They opposed council entry and emphasised Gandhi's constructive programme, including:
 - Promotion of khadi
 - National education
 - Hindu–Muslim unity
 - Removal of untouchability
 - Village reconstruction

Q 122. Ans. C

Explanation:

- **Statement 1 is correct:** The Swaraj Party was founded in January 1923 (the decision was taken in late 1922 after the Gaya Session of Congress) by Chittaranjan Das and Motilal Nehru. The party emerged following the withdrawal of the Non-Cooperation Movement and differences within the Congress regarding participation in legislative councils.
- **Statement 2 is correct:** The Swarajists accepted the entire Congress programme and remained within the Congress organisation. Their disagreement with the No-Changers was only on one issue: entry into legislative councils. They believed that elected representatives should enter the councils and obstruct colonial governance from within, exposing the limitations of constitutional reforms and demanding self-government.

Q 123. Ans. D

Explanation:

- The Responsivist Group emerged within the Swaraj Party during the mid-1920s. It was led by Madan Mohan Malaviya, Lala Lajpat Rai, and N.C. Kelkar.
- **The Responsivists advocated:**
 - ✓ Cooperation with the colonial government in certain circumstances.
 - ✓ Protection of Hindu interests.
 - ✓ Participation in legislatures and local bodies while responding to government initiatives when beneficial.
- This position differed from the original Swarajist strategy of uniform obstruction within the legislative councils

Q 124. Ans. A

Explanation:

- **1939:** Subhas Chandra Bose resigned from the Congress presidency after differences with the Congress leadership and formed the Forward Bloc in June 1939.
- **March 1940:** The Lahore Resolution was passed by the Muslim League, demanding separate states for Muslims.
- **August 1940:** The British Government announced the August Offer to secure Indian cooperation during World War II.
- **January 1941:** Bose escaped from house arrest in Calcutta and later reached Germany, eventually contributing to the formation and leadership of the Indian National Army (INA).

Q 125. Ans. D

Explanation:

- **Radcliffe Boundary Commission — Lord Mountbatten**
 - ✓ The Radcliffe Boundary Commission (1947) was appointed during the tenure of Lord Mountbatten, the last Viceroy of India, to demarcate the boundaries between India and Pakistan.
- **Interim Government — Lord Wavell**
 - ✓ The Interim Government of India was formed on 2 September 1946 under the Wavell Plan during the viceroyalty of Lord Wavell.
- **All India Kisan Sabha — Lord Willingdon**
 - ✓ The All India Kisan Sabha (AIKS) was established in 1936 at Lucknow during the tenure of Lord Willingdon (1931–1936).
- **Butler Commission — Lord Irwin**
 - ✓ The Indian States Committee (Butler Commission) was appointed in 1927 during the viceroyalty of Lord Irwin (1926–1931) to examine the relationship between the princely states and the British Crown.

Q 126. Ans. A

Explanation:

- **Statement 1 is correct:** The militant nationalists (Extremists) clearly declared that Swaraj (self-government/independence) was the ultimate goal of the national movement. Leaders such as Bal Gangadhar Tilak, Bipin Chandra Pal, Lala Lajpat Rai, and Aurobindo Ghose rejected the moderate demand for limited reforms and emphasized complete self-rule.

- **Statement 2 is incorrect :** The militant nationalists did not believe that India's progress depended on continued British guidance. On the contrary, they argued that foreign rule was the chief obstacle to India's development and firmly rejected the idea that India could prosper under British control. They advocated self-reliance (Atmasakti), national education, Swadeshi, and mass political action.

Q 127. Ans. D

Explanation:

- The agitation against the Partition of Bengal (1905) was a broad-based mass movement that drew support from various sections of society.
- **Zamindars:**
 - ✓ Many zamindars actively opposed the partition, viewing it as an attempt to weaken Bengali unity and nationalism.
- **Students:**
 - ✓ Students played a crucial role in organizing meetings, processions, picketing foreign goods, and spreading the message of Swadeshi and Boycott.
- **Merchants:**
 - ✓ Merchants and traders supported the movement by promoting indigenous goods and participating in the boycott of British products.
- **Women:**
 - ✓ For the first time on a significant scale, women participated in nationalist activities, attended public meetings, joined processions, and supported the Swadeshi movement.

Q 128. Ans. B

Explanation:

- **The famous statement:**
 - ✓ "Bengal united is a power; Bengal divided will pull in several different ways." was made by Sir Herbert Hope Risley (H.H. Risley), who was the Home Secretary to the Government of India during the time of the Partition of Bengal (1905).
 - ✓ In an official note dated 6 December 1904, Risley openly acknowledged that one of the objectives of the partition was to weaken the growing nationalist movement in Bengal by dividing the Bengali-speaking population.
 - ✓ **He further wrote:**
 - "One of our main objects is to split up and thereby to weaken a solid body of opponents to our rule."

Q 129. Ans. C

Explanation:

- **Statement 1 is correct:** As part of the Swadeshi Movement, nationalists sought to develop an education system free from colonial influence. Consequently, the National Council of Education (NCE) was established on 15 August 1906 in Bengal to promote national education and scientific and technical learning.
- **Statement 2 is correct:** A National College was established in Calcutta under the auspices of the National Council of Education, and Aurobindo Ghose was appointed as its first Principal. He used the institution to inspire nationalist ideals among students.

Q 130. Ans. B

Explanation:

- **March 1940:** The Muslim League adopted the Lahore Resolution, demanding separate states for Muslims.
- **May 1940:** Winston Churchill became Prime Minister of Britain during World War II.
- **August 1942:** The Congress passed the Quit India Resolution at the Bombay session, launching the Quit India Movement.
- **1944:** At its Karachi session, the Muslim League raised the slogan "Divide and Quit" in response to the Congress demand of "Quit India."

Q 131. Ans. B

Explanation:

- Under the SEZ Act, 2005 (Section 2(i)), the Domestic Tariff Area means the whole of India, including the territorial waters and continental shelf, but excluding the areas of the Special Economic Zones. So the DTA is all national territory minus the SEZ enclaves.

Q 132. Ans. C

Explanation:

- The Index of Eight Core Industries (ICI) is compiled with 2011–12 as the base year (2011–12 = 100). It is released monthly by the Office of Economic Adviser under the Department for Promotion of Industry and Internal Trade.
- **The index tracks the performance of eight infrastructure industries:**
 - ✓ Coal
 - ✓ Crude Oil
 - ✓ Natural Gas
 - ✓ Refinery Products
 - ✓ Fertilizers
 - ✓ Steel
 - ✓ Cement
 - ✓ Electricity

- These industries collectively account for 40.27% of the weight in the Index of Industrial Production (IIP).

● Highest and Lowest Weightage in ICI

Core Industry	Weight (%)
✓ Refinery Products	28.04 (Highest)
✓ Steel	17.92
✓ Electricity	19.85
✓ Coal	10.33
✓ Cement	5.37
✓ Natural Gas	6.88
✓ Crude Oil	8.98
✓ Fertilizers	2.63 (Lowest)

Q 133. Ans. D

Explanation:

- The economy is generally classified into Primary, Secondary, and Tertiary sectors based on the nature of economic activities.
- The Tertiary Sector (Service Sector) includes activities that provide services rather than producing tangible goods. Banking, insurance, and transport are classic examples of service activities because they facilitate production, trade, and consumption without directly creating physical products.

Q 134. Ans. B

Explanation:

- **V.K.R.V. Rao:** He is credited with making the first scientific estimate of India's national income for the year 1931–32. He divided the Indian economy into two parts (agricultural and corporate sectors) and utilized a combination of the census of output and census of income methods.

Q 135. Ans. A

Explanation:

- **Statement 1 is correct.** The Union Budget 2026–27 has proposed setting up AVGC Content Creator Labs (CCLs) in 15,000 secondary schools and 500 colleges across India, with a financial allocation of ₹250 crore under the Ministry of Information and Broadcasting. These labs aim to build foundational skills in animation, visual effects, gaming, and content creation among students, and will be

rolled out through the Indian Institute of Creative Technologies (IICT), Mumbai, which has been designated as the nodal agency for this initiative.

- **Statement 2 is incorrect.** India does not have the largest anime fan base in the world; it currently holds the second-largest position globally, as per the FICCI-EY 2024 report.
- **Statement 3 is incorrect.** The National Centre of Excellence (NCoE) for Animation, Visual Effects, Gaming, Comics, and Extended Reality (AVGC-XR) has been established in **Mumbai**, not Hyderabad. It currently operates from the IICT-National Film Development Corporation (NFDC) campus in Mumbai. This NCoE is being developed as a hub-and-spoke model institution, similar in stature to premier institutions like IITs and IIMs, to build a world-class talent pool for India's creative economy.

Q 136. Ans. B

Explanation:

- **Statement 1 is correct:** The Mahatma Gandhi Gram Swaraj initiative explicitly proposes to build rural self-reliance by strengthening traditional village industries, specifically focusing on the khadi, handloom, and handicraft sectors.
- **Statement 2 is correct:** A core pillar of the scheme is to modernize these traditional sectors. It streamlines training and skilling to improve the quality of processes and production, with the ultimate goal of achieving global market linkages and building a strong "Brand India" for handmade goods.
- **Statement 3 is correct:** The MGGSI aims to create vital backward and forward linkages across the value chains of khadi, handlooms, and handicrafts. This includes local sourcing mechanisms that integrate directly with the **One District One Product (ODOP)** initiative to support local weavers and artisans.
- **Statement 4 is incorrect:** While Design Resource Centres (DRCs) are a crucial part of building design-oriented excellence in the handloom sector, the government has established **16 Design Resource Centres** across the country (such as the one in Hyderabad, Telangana), not in every state.

Q 137. Ans. C

Explanation:

- The Cash Reserve Ratio (CRR) is the percentage of a bank's Net Demand and Time Liabilities (NDTL)

that must be maintained as cash reserves with the Reserve Bank of India (RBI).

- Banks cannot use these funds for lending or investment, making CRR an important tool for controlling liquidity in the economy. When the RBI increases CRR, the lending capacity of banks decreases; when it reduces CRR, more funds become available for lending.

Q 138. Ans. D

Explanation:

- **The Union Budget 2026–27 proposes seven High-Speed Rail Corridors as 'Growth Connectors' to promote environmentally sustainable passenger transport. These are:**
 - ✓ Mumbai – Pune
 - ✓ Pune – Hyderabad
 - ✓ Hyderabad – Bengaluru
 - ✓ Hyderabad – Chennai
 - ✓ Chennai – Bengaluru
 - ✓ Delhi – Varanasi
 - ✓ Varanasi – Siliguri

Q 139. Ans. C

Explanation:

- As per the Economic Survey 2025-26, under the Special Assistance to States for Capital Investment (SASCI) scheme, the Union Government incentivised States to maintain capital expenditure at around **2.4 percent** of GDP in FY 2024–25, with the objective of sustaining public investment and supporting long-term economic growth.

Q 140. Ans. C

Explanation:

- According to the Economic Survey 2025–26, 5G services are available in 99.9 per cent of districts across the country.
- The Survey also notes that India's tele-density has reached 86.76 per cent, reflecting significant expansion in digital connectivity and telecommunications infrastructure. Hence, 86.76 per cent refers to tele-density, not 5G coverage.

Q 141. Ans. B

There are two series merged in the given series alternatively. The first series starts with 2 and the next number is twice the previous number. And the second series starts with 3 and the next number is thrice the previous number.

$$\begin{array}{ccccccc}
 \lceil \times 3 & \top \times 3 & \top \times 3 & \rceil \\
 2, & 3, & 4, & 9, & 8, & 27, & 16, & ? \\
 \lfloor \times 2 & \bot \times 2 & \bot \times 2 & \rfloor
 \end{array}$$

$$2 \times 2 = 4$$

$$4 \times 2 = 8$$

$$8 \times 2 = 16$$

And

$$3 \times 3 = 9$$

$$9 \times 3 = 27$$

$$27 \times 3 = 81$$

Q 142. Ans. B

- The deficiency of Vitamin A causes Night blindness. Similarly, the deficiency of Vitamin B12 causes Anemia.

Q 143. Ans. C

'BAIT' can be formed by the letters of the word 'CALIBRATION'.

'CRAB' can be formed by the letters of word 'CALIBRATION'

'TRAIT' cannot be formed by the letters of the word 'CALIBRATION' as there is only one 'T' in the given word.

'BRAIN' can be formed by the letters of word 'CALIBRATION'

Q 144. Ans. A

The number of odd days in a normal year is 1 and the number of odd days in leap years are 2.

There are 3 normal years (2025, 2026 and 2027) and 2 leap year 2024 and 2028 from July 5, 2023 to July 5 2028

So, number of odd days will be:

From July 5, 2023 to July 4, 2024: 2

From July 5, 2024 to July 4, 2025: 1

From July 5, 2025 to July 4, 2026: 1

From July 5, 2026 to July 4, 2027: 1

From July 5, 2027 to July 4, 2028: 2

So, the total number of odd days is 7 or 0.

So, on July 5, 2028, it will be Wednesday

Q 145. Ans. B

The logic followed here is, the center number is the sum of all the numbers given around it and then multiply the obtained sum by the lowest number given.

In 1st figure,

$$(23 + 6 + 17 + 12) \times 6 = (58) \times 6 = 348$$

In 3rd figure,

$$(15 + 26 + 19 + 7) \times 7 = (67) \times 7 = 469$$

Similarly, in 2nd box,

$$(16 + 11 + 8 + 24) \times 8 = (59) \times 8 = 472$$

Q 146. Ans. B

The profit-sharing ratio between A and B will be the same as the ratio of their investments.

$$A : B = 18,000 : 30,000 = 3 : 5$$

$$\text{Total Profit} = ₹4,320$$

$$A's \text{ Share of Profit} = 3/8 \times 4,320 = ₹1,620$$

$$B's \text{ Share of Profit} = 5/8 \times 4,320 = ₹2,700$$

Now, A has given ₹480 to B.

$$\text{Therefore, the amount B now possesses} = 2,700 + 480 = ₹3,180$$

Q 147. Ans. B

$$A \times (x + 12)\% - A \times x\% = 24$$

$$A \times 12\% = 24$$

$$A = 200$$

$$(A + 20)\% \text{ of } (2x - 3) = 99$$

$$(200 + 20) \times (2x - 3)/100 = 99$$

$$11/5 \times (2x - 3) = 99$$

$$(2x - 3) = 45$$

$$x = 24$$

$$10x = 240$$

$$A = 200$$

$$\text{Percentage} = 240/200 \times 100\% = 120\%$$

Q 148. Ans. C

$$\text{Total Cost} = ₹18,000$$

$$\text{Total Quantity} = 600 \text{ kg}$$

$$\text{Spoiled Wheat} = 600 \times 25\% = 150 \text{ kg}$$

$$\text{Remaining Wheat} = 600 - 150 = 450 \text{ kg}$$

$$\text{Selling Price for a 20\% Profit} = 18,000 \times 1.2 = ₹21,600$$

$$\text{Selling Price per kg} = 21,600 / 450 = ₹48$$

Therefore, the required selling price = ₹48/kg.

Q 149. Ans. A

According to the question:

$$\text{Time} = (\text{February 17} - \text{April 30}), 2024 = 73 \text{ days}$$

$$R = 8$$

$$\text{Simple Interest} = (P \times R \times T) / (100 \times 365)$$

$$\Rightarrow ₹15,000 \times 8 \times 73 / (100 \times 365)$$

$$\Rightarrow ₹240$$

Q 150. Ans. A

Let the first number be A and the second be B.

According to the question, 70% of A = (1/2) of B

$$\Rightarrow 70/100 \times A = 1/2 \times B$$

$$\Rightarrow 7/10 \times A = 1/2 \times B$$

$$\Rightarrow A : B = 1/2 : 7/10$$

$$\Rightarrow A : B = 10 : 14 = 5 : 7$$

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